

AKADEMIKA

RFL-SCT

**SATELLITE COMMUNICATION
TRAINER**

EXPERIMENTAL MANUAL

.... A MESSAGE FROM



Today's system designers are faced with tomorrow's problems. SATELLITE COMMUNICATION is one of the important subject need to teach while learning electronics. It is our vision to provide you with the product you need for training ensuring lasting reliability & quality.

Company's Motto

- Light years ahead – refers to leadership.

We strive to be the best in what we do and maintain high standards in the areas of

• Design • Quality • Value • Delivery • Support

We are indeed light years ahead of the competition in this field. We guarantee our valued customers great satisfaction.

As you will move through this manual you will quickly discover that we have complete, truly innovative & superior training products. We are so committed to quality that we back our products with a complete and comprehensive warranty.

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SAFETY RULES

- Read carefully and follow the instructions mentioned in this manual. This user manual includes all the important points about the installation, use and the maintenance of the product. Keep this manual always with you, for quick reference.
- After unpacking the product, arrange all the accessories in proper order, so that their integrity is checked with the packing list. Also, ensure that the accessories have no visible damage.
- Before connecting the power supply to the kit, be sure that the jumpers and the connecting chords are connected correctly, as per the experiment.
- In WL-SCT, signals are to be monitored with an oscilloscope or with the help of indicators as explained in the manual. For observation of signals on oscilloscope, either use X10 (Attenuation Probe) or 180Ω resistance in series with normal oscilloscope probe.
- This kit must be employed only for the use for which it has been conceived, i.e. as educational kit and must be used under the direct survey of expert personnel. Any other use is inadvisable and dangerous too. The manufacturer cannot be considered responsible for eventual damages due to improper, wrong or unreasonable uses.
- In case of any fault or malfunctioning in the trainer kit, turn off the power supply. Please do not tamper the kit. If you require our service, kindly contact the service centre for technical assistance.
- The kits are liable to malfunction/under-perform if they are not operated under standard environmental conditions of temperature and humidity.

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This kit is warranted against defects in workmanship and materials. Any failure due to defect in either workmanship or material should occur under normal use within a year from the original date of purchase, such failure will be corrected free of charge to the purchaser by repair or replacement of defective part or parts. When the failure is result of user's neglect, natural disaster or accident, we would charge for repairs, regardless of the warranty period. The warranty does not cover include perishable items like connecting chords, crystals, etc. and other imported items.

Conditions and Limitations

The warranty is void and inapplicable if the defective product is not brought or sent to our authorized service center or sales outlet within the warranty period. Defective product will be Akademia Lab Solutions' sole judgment. The defective product will be replaced with a new one or repaired, without charge or with charge.

In the warranty period if the service is needed, the purchaser should get in touch with the service center or the sales outlet. The purchaser should return the product to the service center or the sales outlet at his or her sole expense. The loss and damage in transit will be outside the preview of this warranty. A returned product must be accompanied by a written description of the defects. Type and Model No. of the kit has to be mentioned specifically. We return the product to the purchaser at our expense. In case the warranty does not cover the product on Akademia Lab Solution's judgment, we would repair the product after obtaining prior permission from purchaser who would receive an estimate statement about the repairing charges. In such cases, Akademia Lab Solutions bares the transporting expenses required to send back all the repaired products for the moment, and then repairs and transporting expenses will be charged against the purchaser by the statement of accounts.

When the authorized sales agents sell our products, they must notify the purchaser of the warranty contents, but they have no rights to stretch the meaning of original warranty contents or to offer an additional warranty. Akademia Lab Solutions does not provide any other promise or suggestive warranty and hold no liability for the damage caused by negligence, abnormal use or natural disaster. Akademia Lab Solutions is not responsible for the damages even if it is notified of above dangers in advance as well.

For more special service or overall repairs, maintenance and up gradation of products, be sure to contact our service center or the sales outlet.

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TECHNICAL SPECIFICATIONS

- Frequency : 2.4GHz, 2.427GHz, 2.454GHz, 2.481GHz
- Display : 16 X 2 LCD
- Spurious output : -30dB typical
- RF output impedance : 75 ohms unbalanced
- Modulation : Frequency modulation of audio and video signal
- Audio subcarrier : 5 and 5.5 MHz
- Video subcarrier : 8 MHz maximum
- RF carrier o/p level : +3dBm nominal
- Path loss : Up to 20dB variable control
- LF Modulation I/P : Internal 1KHz sine wave /
- External mic or External Analog input
- HF Modulation I/P : 5MHz Bandwidth typical
- Analog : 5Vp/p maximum
- Digital : TTL Compatible
- RS232 : PC serial port compatible
- Input Power Supply : 220V AC $\pm$ 10%

Deliverables

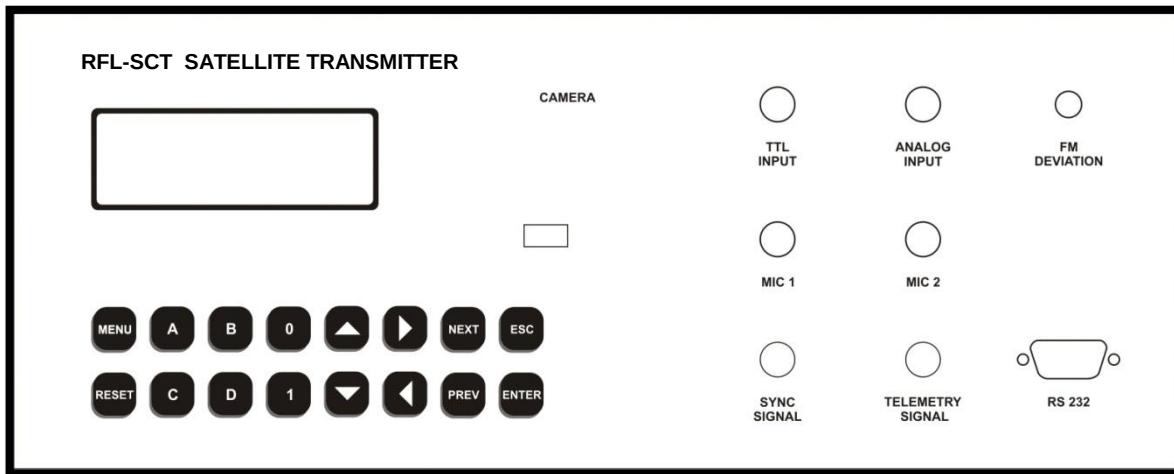
RFL-SCT		Quantity
1	Transmitter	1
2	Receiver	1
3	Emulator	1
4	Power cord	3
5	Monopole Antenna	4
6	MIC with connecting wire	1
7	TV SET WITH ACCESSORISE	1
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FUNCTIONAL BLOCK



INTRODUCTION

Satellite Uplink Transmitter



Display: This is a 20X4 alphanumeric Backlit LCD used to display the functions of operation of the transmitter.

Camera: For capturing video image.

Keypad: 8X2 matrix keypad for performing various operations.

MENU Key for selecting from available menus.

RESET Key for power reset for the instrument.

A, B, C, and D, 0, 1 Alphanumeric keys for providing required input/data to the instrument.

▲ Upward key to move the cursor upward.

▼ Downward key to move the cursor downward.

▶ Forward arrow key to move the menu options in forward direction.

◀ Backward arrow key to move the menu options in backward direction.

ESC Escape key to move out from the current message and go to the previous message

ENTER Enter key to set the selected option.

TTL INPUT: BNC connector input for TTL signal.

ANALOG INPUT: Input for Analog signal with BNC connector.

FM DEVIATION: Potentiometer provided for inserting FM deviation in Audio Signal.

MIC 1: Socket for connecting external Mike for Audio Signal.

MIC 2: Socket for connecting external Mike for Audio Signal.

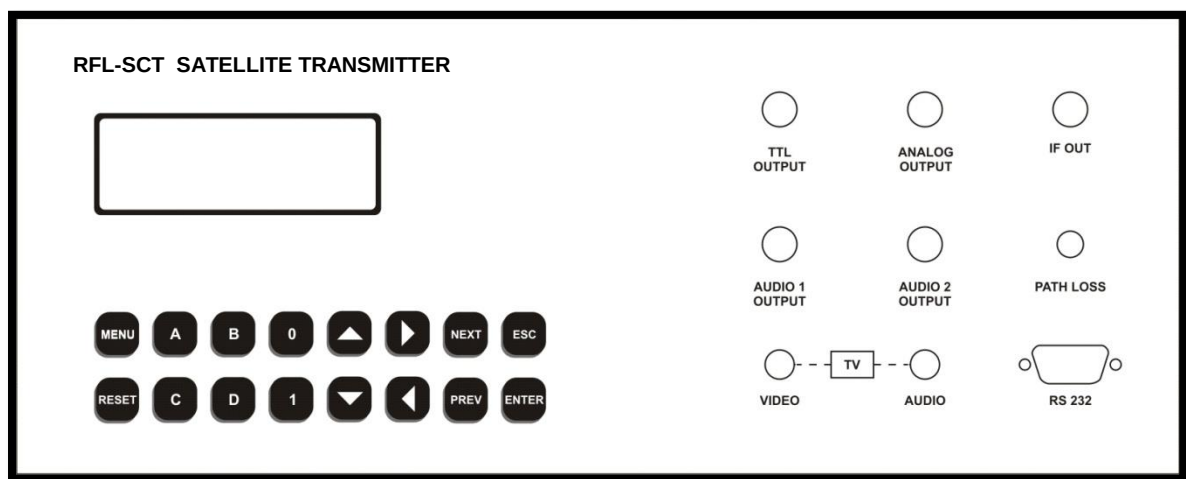
SYNC SIGNAL: BNC output connector for observing SYNC pulse in video signal,

TELEMETRY SIGNAL: BNC output connector for observing Encoded data for telemetry.

RS 232: D type connector for RS232 interface.

Note: At the Back panel of the Satellite Uplink Transmitter, SMA connector is provided for Uplink Transmitter Antenna.

Satellite Downlink Receiver



Display: This is a 20X4 alphanumeric Backlit LCD used to display the functions of operation of the transmitter.

Keypad: 8X2 matrix keypad for performing various operations.

MENU Key for selecting from available menus.

RESET Key for power reset for the instrument.

A, B, C, and D, 0, 1 Alphanumeric keys for providing required input/data to the instrument.

▲ Upward key to move the cursor upward.

▼ Downward key to move the cursor downward.

▶ Forward arrow key to move the menu options in forward direction.

◀ Backward arrow key to move the menu options in backward direction.

ESC: Escape key to move out from the current message and go to the previous message.

ENTER: Enter key to set the selected option.

TTL OUTPUT: BNC connector Output for TTL signal.

ANALOG OUTPUT: BNC connector Output for Analog signal.

IF OUT: SMA connector Output for IF signal.

AUDIO 1 OUTPUT: BNC Connector output for Audio Signal.

AUDIO 2 OUTPUT: BNC Connector output for Audio Signal.

PATH LOSS: Potentiometer for adjusting the Path loss (i.e. level of attenuation for transmitted signal).

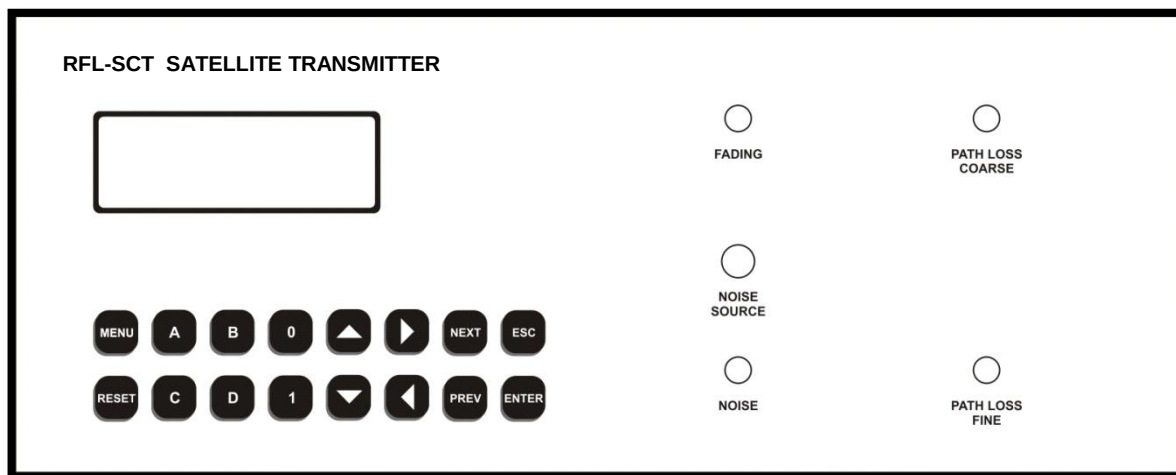
VIDEO: Connector for Video output.

AUDIO: Connector for Audio output.

RS 232: D type interface for RS232 interface.

Note: At the Back panel of the Satellite Downlink Receiver, SMA connector is provided for Downlink Receiver Antenna.

Satellite Link Emulator



Display: This is a 20X4 alphanumeric Backlit LCD used to display the functions of operation of the transmitter.

Keypad: 8X2 matrix keypad for performing various operations.

MENU Key for selecting from available menus.

RESET Key for power reset for the instrument.

A, B, C, and D, 0, 1 Alphanumeric keys for providing required input/data to the instrument.

▲ Upward key to move the cursor upward.

▼ Downward key to move the cursor downward.

▶ Forward arrow key to move the menu options in forward direction.

◀ Backward arrow key to move the menu options in backward direction.

ESC: Escape key to move out from the current message and go to the previous message.

ENTER: Enter key to set the selected option.

NOISE: Potentiometer for adjusting amount of noise to be added.

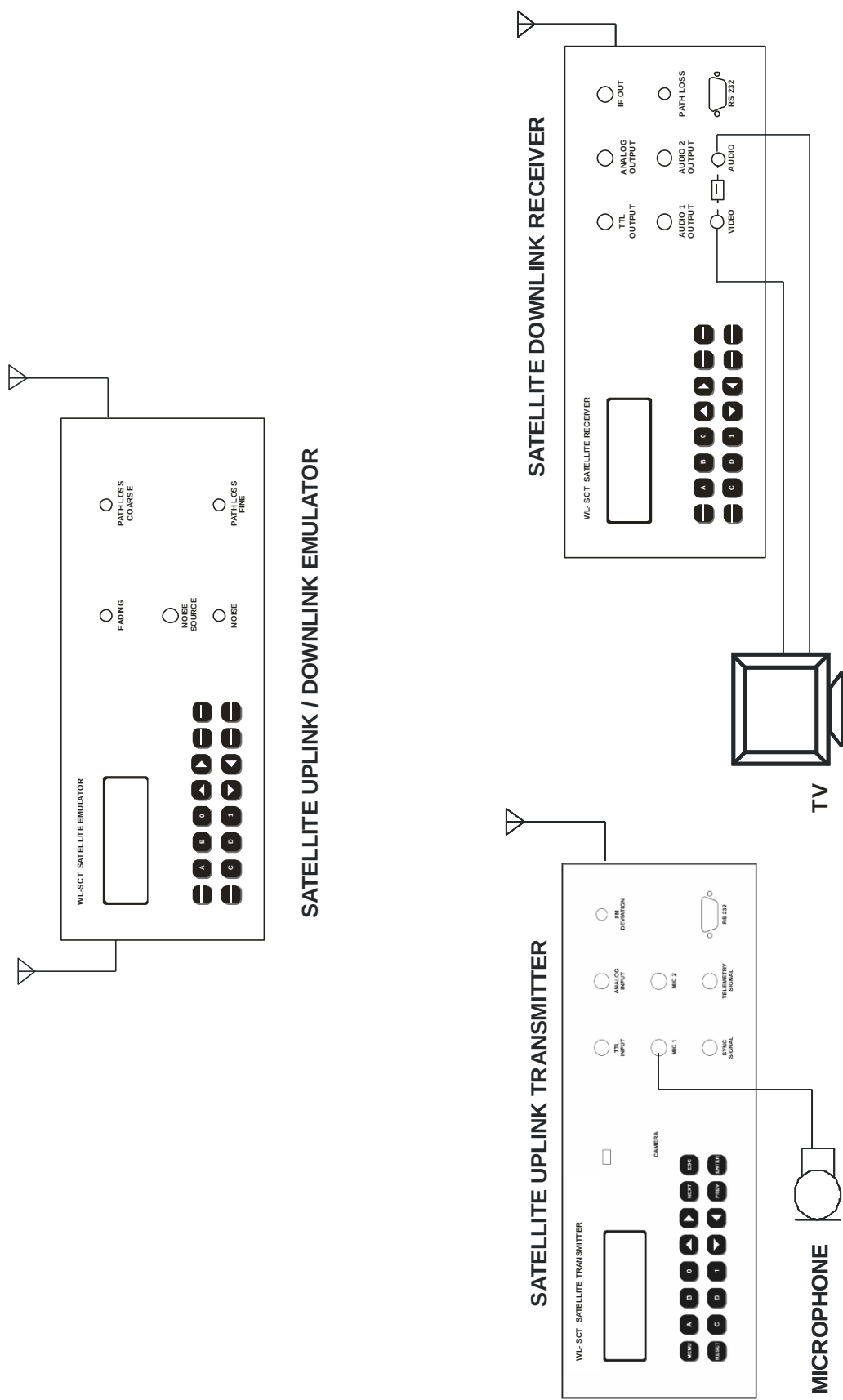
PATH LOSS: Potentiometer for adjusting the Path loss (i.e. level of attenuation for transmitted signal).

FADING: Potentiometer for adjusting fading of signal

Noise Source: The BNC terminal to observe noise

Note: At the Back panel of the Satellite Link Emulator, two SMA connectors are provided for Uplink and Downlink Antenna.

EXPERIMENT NO. 1



Block diagram for Expt.1 ACTIVE / PASSIVE SATELLITES UPLINK / DOWNLINK & TRANSPONDERS

EXPERIMENT NO: 1

NAME

Active/Passive Satellites, Uplink/Downlink & Transponders

OBJECTIVE

- To set up an active & passive satellite communication link and study their difference.
- To study the advantages of satellite communication.
- To study the communication satellite link design: process of transmitting a signal to a satellite (UPLINKING), reception of same signal via satellite (DOWN LINKING) and functioning of transponder of a satellite.

EQUIPMENTS

- Satellite uplink transmitter, satellite downlink receiver and satellite link emulator
- Antennas
- Microphone, 10 MHz Function Generator, Digital Oscilloscope.

THEORY

The UPLINK

In uplink station, the signals have to be sent at a differing frequency, usually in the higher 14 GHz band, to avoid interference with downlink signals. Another function performed by the uplink station is to control tightly the internal functions of the satellite itself (such as station keeping accuracy). Uplinks are controlled so that the transmitted microwave power beam is extremely narrow, in order not to interfere with adjacent satellites in the geo-arc. The powers involved are several hundred watts.

The transmitter power for earth station is provided by high power amplifiers. The large power can be supplied to these amplifiers. The transmitting antenna and amplifier units are placed on the ground therefore there is no limitation on size, weight etc. parameters. Therefore high effective isotropic radiated power (EIRP) levels are possible for satellite uplinks. The power levels of 40-60 dB W are possible even at high frequency bands like K-bands and V-bands. The beam pattern of the satellite decides the power actually sent to the satellite and interference to the neighboring satellite.

As the beam becomes narrower from the earth station, the interference is reduced, but it should track the satellite location exactly. Also the gain of the earth station is increased. Therefore as the beam width is narrowed, the satellite pointing should be improved. This allows the satellite to be placed closer in the same orbit. As the uplink carrier frequency goes on increasing, the size of Antenna goes on reducing. This reduces the size of complete earth station.

THE TRANSPONDERS

Each satellite has a number of transponders with access to a pair of receive/transmit antennas and associated electronics for each channel. For example, in Europe, the uplink sends signals at a frequency of about 14 GHz; these are received, down-converted in frequency to about 11/12 GHz and boosted by high power amplifiers for re-transmission to

earth. Separate transponders are used for each channel and are powered by solar panels with back up batteries for eclipse protection. The higher the power of each transponder, the fewer channels will be possible with a given number of solar panels, which in turn, is restricted by the maximum payload of launch vehicles as well as cost. Typical power consumption for a satellite such as ASTRA 1A is 2.31 kW with an expected lifetime of 12.4 years. Satellites are conveniently categorized into the following three power ranges:

The satellite Transponder receives the uplink transmission from the earth station and retransmits the signal on downlink.

The uplink transmission is received by the antenna of the satellite. Through diplexer it is given to the front end receiver.

The front end receiver increases the signal to noise ratio of the signal received and provides amplification. The power received at the antenna of satellite via uplink is very small. Therefore front end receiver provides amplification to the signal. Carrier processing involves the demodulation of the uplink carrier frequencies and demodulation of the information on downlink frequencies. It can also change the modulation format for downlink.

Normally uplink and downlink frequencies are separate. This is done so that uplink and downlink frequencies should not mix with each other. Therefore same antenna is used for the transmission of downlink frequencies. The diplexer performs the job of simultaneous transmission and reception through the antenna. Since the uplink and downlink frequency bands are separate, simultaneous reception and transmission has no problem.

The power amplifier is provided in the transponder to increase the power level of demodulated downlink carrier. The power level is such that it should reach satisfactorily to the earth stations. The gain of the typical transponder is around 80-100dB.

1. Low power: These have transponder powers around the 20 W marks and are primarily general telecommunication satellites. Due to the low transmission power of each transponder they can support many channels with the available collected solar energy. Many of these transponders relay program material for cable TV operators but, unfortunately, receiving dishes of monstrous proportions are necessary for noise free reception, often in excess of 1 meter. Even so, domestic TV reception is not the primary reason for the existence of such high channel capacity satellites. Transponder bandwidths can vary.

2. Medium power: These satellites have typical transponder powers of around 45 W, such as those on board Astra 1A. Such satellites are now commonly termed semi-DBS (direct broadcast service) and represent the first serious attempt to gain public approval by offering the prospect of dustbin-Lid-sized dishes of 60 cm diameter. About sixteen transponders are average for this class at the present time. Medium power satellites usually operate in the frequency band 10.95 GHz to 11.70 GHz and form the fixed satellite service (FSS). The transponder bandwidths are commonly 27 MHz or 36 MHz. Some medium power satellites, such as the Eutelsat II series, also have a number of transponders that can be active in the 12.5 GHz to 12.75 GHz band.

3. High power: These pure DBS satellites have transponder powers exceeding 100 W and have a correspondingly reduced channel capacity of around four perhaps five channels. The specified dish size is minimal, about 30 to 45 cm in the central service area. European transponder frequencies are in the band 11.70 to 12.50 GHz which is known as the DBS band. It has been agreed that the **transponder bandwidths are 27 MHz.**

The DOWNLINK

The medium used to transmit signals from satellite to earth is microwave electromagnetic radiation which is much higher in frequency than normal broadcast TV signals in the

VHF/UHF bands. **Microwaves still exhibit a wave-like nature but inherit a tendency to severe attenuation by water vapors or any obstruction in the line of sight of the antenna.** The transmitted microwave power is extremely weak by the time it reaches earth and unless well designed equipment is used, and certain installation precautions are taken, the background noise can ruin the signal. Televisions receive only **(TVRO)** site consists of an antenna designed to collect and concentrate the signal to its focus where a feed horn is precisely located. This channels microwave to an electronic component called a low noise block **(LNB)**, which amplifies and down-converts the signal to a more manageable frequency for onward transmission, by cable, to the receiver located inside the dwelling.

The amplifier and transmitting antennas now are placed on the satellite itself for downlink. This limits the size and weight of the transmitting antennas and complete amplifier.

The power at the satellite is limited. Therefore small power can be transmitted from the satellite on downlink. The power output from the satellites on the downlink depends on the downlink frequencies. The downlink frequencies are lower than uplink frequencies. The requirements of downlink frequencies are that,

The attenuation should be less compared to the uplink frequencies because the power available at the satellite transmitter is limited.

For the same transmitted power, the low frequencies travel more compared to high frequencies.

To fulfill these requirements low frequencies are used for downlinks compared to uplink frequencies. The beams of downlink frequencies are designed such that they provide the required coverage area. The EIRP of the satellite or receiver gain does not directly affect the downlink quality. The choice of downlink frequency depends on the maximum power that can be transmitted and atmospheric losses.

PROCEDURE

- Connect the Satellite uplink transmitter to AC mains outlet with the lead provided.
- Switch ON the transmitter and the Welcome Message will be displayed for 5Seconds.

*	*	*	*	*	*	*	W	E	L	C	O	M	E	*	*	*	*	*	*
									T	O									
						A	K	A	D	E	M	I	K	A					

- After the welcome message, another message for Menu will be displayed.

	S	A	T	E	L	L	I	T	E		T	R	A	I	N	I	N	G	
							S	Y	S	T	E	M							
			P	R	E	S	S		M	E	N	U		K	E	Y			

- Press **MENU** key on the front panel of the Satellite uplink transmitter. The message for selection from the menu options will be displayed.

M	E	N	U	:	S	E	L	E	C	T		O	P	T	I	O	N	S	
A	)		U	P	L	I	N	K		F	R	E	Q		B	A	N	D	
B	)		I	N	P	U	T		C	H	A	N	N	E	L	S			
C	)		V	I	E	W		S	E	T	T	I	N	G	S				

- Press **A** key to select the Uplink Frequency Band. The message for available frequencies to be selected will be displayed.

		U	P	L	I	N	K		F	R	E	Q		B	A	N	D		
				(	A	L	L		I	N		M	H	z	)				
A	)		2	4	0	0			B	)		2	4	2	7				
C	)		2	4	5	4			D	)		2	4	8	1				

- The transmitting frequency from 2.400GHz, 2.427GHz, 2.454 GHz, 2.481 GHz can be selected by means of pressing a corresponding key (i.e. A, B, C, D) provided on the front panel. This indicates that each channel is spaced 27 MHz apart.
- All frequencies are PLL locked. PLL means that when both receiver and transmitter are set at same frequency, they are accurate to less than 10 KHz of each other and no further tuning and repeated adjustments are required.
- Now bring the transmitter to 2.481 GHz by pressing key **D**. The message for selected 2.481 GHz frequency band will be displayed for 5 seconds.

		U	P	L	I	N	K		F	R	E	Q		B	A	N	D		
2	4	8	1		M	H	Z		I	S		S	E	L	E	C	T	E	D

- Now press ECS key to go to the main menu.

M	E	N	U	:	S	E	L	E	C	T		O	P	T	I	O	N	S	
A	)		U	P	L	I	N	K		F	R	E	Q		B	A	N	D	
B	)		I	N	P	U	T		C	H	A	N	N	E	L	S			
C	)		V	I	E	W		S	E	T	T	I	N	G	S				

- Press key **B** on the front panel of the transmitter. The message for the Input Channels will be displayed.

				I	N	P	U	T		C	H	A	N	N	E	L	S		
A	U	D	I	O		C	H	1	:	O	F	F							
A	U	D	I	O		C	H	2	:	O	F	F							
V	I	D	E	O		C	H	3	:	O	F	F							

- See that the cursor is in front of the AUDIO CH1: Use **▲** UP arrow or **▼** DOWN arrow keys to do that. Use forward arrow **▶** or down arrow key **◀** to select AUDIO CH1 at MIC1 and Video CH 3 at VIDEO.

				I	N	P	U	T		C	H	A	N	N	E	L	S		
A	U	D	I	O		C	H	1	:	M	I	C	1						
A	U	D	I	O		C	H	2	:	O	F	F							
V	I	D	E	O		C	H	3	:	V	I	D	E	O					

Press **ENTER** to set it for MIC1 and CH3 to VIDEO.

M	E	N	U	:	S	E	L	E	C	T		O	P	T	I	O	N	S	
A	)		U	P	L	I	N	K		F	R	E	Q		B	A	N	D	
B	)		I	N	P	U	T		C	H	A	N	N	E	L	S			
C	)		V	I	E	W		S	E	T	T	I	N	G	S				

- Connect the microphone to the MIC 1 post of the UPLINK TRANSMITTER. Make sure that the FM DEVIATION potentiometer is at the fully anticlockwise position.
- Connect the antenna with a SMA lead to R.F. out of Transmitter.
- Connect the Satellite EMULATOR to AC mains outlet with the lead provided.
- Switch ON the Satellite EMULATOR and the Welcome Message will be displayed for 5Seconds.

*	*	*	*	*	*	*	W	E	L	C	O	M	E	*	*	*	*	*	*
									T	O									
							A	K	A	D	E	M	I	K	A				

- After the welcome message, another message for Menu will be displayed.

S	A	T	E	L	L	I	T	E	T	R	A	I	N	I	N	G
						S	Y	S	T	E	M					
		P	R	E	S	S		M	E	N	U		K	E	Y	

- Press **Menu** key to go to the menu options.

M	E	N	U	:	S	E	L	E	C	T		O	P	T	I	O	N	S
A	)	U	P	/	D	O	W	N		L	I	N	K		F	R	E	Q
B	)	C	H	A	N	N	E	L		D	A	T	A					

- Press key **A** to go to the menu for uplink/downlink frequency selection for the Emulator.

		U	P	/	D	O	W	N		L	I	N	K		F	R	E	Q	
A	)		U	P	L	I	N	K		F	R	E	Q		B	A	N	D	
B	)	D	O	W	N	L	I	N	K		F	R	E	Q		B	A	N	D

- Press key **A** to go to uplink Frequency Band Selection.

		U	P	L	I	N	K		F	R	E	Q		B	A	N	D	
				(	A	L	L		I	N		M	H	z	)			
A	)		2	4	0	0			B	)		2	4	2	7			
C	)		2	4	5	4			D	)		2	4	8	1			

- Select uplink frequency of 2.481 GHz by pressing key **D**.

		U	P	L	I	N	K		F	R	E	Q		B	A	N	D		
2	4	8	1		M	H	z		I	S		S	E	L	E	C	T	E	D

- This message will be flashed for 5 seconds, Press ESC key to go to previous Menu. Press B key to Downlink Frequency Band selection.

		D	O	W	N	L	I	N	K		F	R	E	Q		B	A	N	D
				(	A	L	L		I	N		M	H	z	)				
A	)		2	4	0	0			B	)		2	4	2	7				
C	)		2	4	5	4			D	)		2	4	8	1				

- Press key **A** to select the downlink frequency of 2.400 GHz. The message for selected downlink frequency of 2.400GHz will be displayed for 5 seconds. Press ECS key two times to go to main MENU.

		D	O	W	N	L	I	N	K		F	R	E	Q		B	A	N	D
--	--	---	---	---	---	---	---	---	---	--	---	---	---	---	--	---	---	---	---

2	4	0	0		M	H	z		I	S		S	E	L	E	C	T	E	D
---	---	---	---	--	---	---	---	--	---	---	--	---	---	---	---	---	---	---	---

A clear sound at the receiver indicates that a microwave satellite communication link has been set up successfully. In active satellites, the frequency is translated by transponders in satellite and then sent back to receiver after amplifying the signal at different frequency.

Whereas in Passive satellite, signal is only reflected back to the receiver and no freq. translation and power amplification takes place. Active satellite uses up external energy (solar or battery) and active circuits to perform the frequency translation and power amplification. Plus WL-SCT is useful where direct line of sight link over long distances is not possible due to curvature of earth.

Up linking in commercial C band is at 5.925 – 6.425 GHz and

Up linking in commercial Ku band is at 14.000 – 14.500 GHz.

Down linking in commercial C band is at 3.700 – 4.200 GHz and

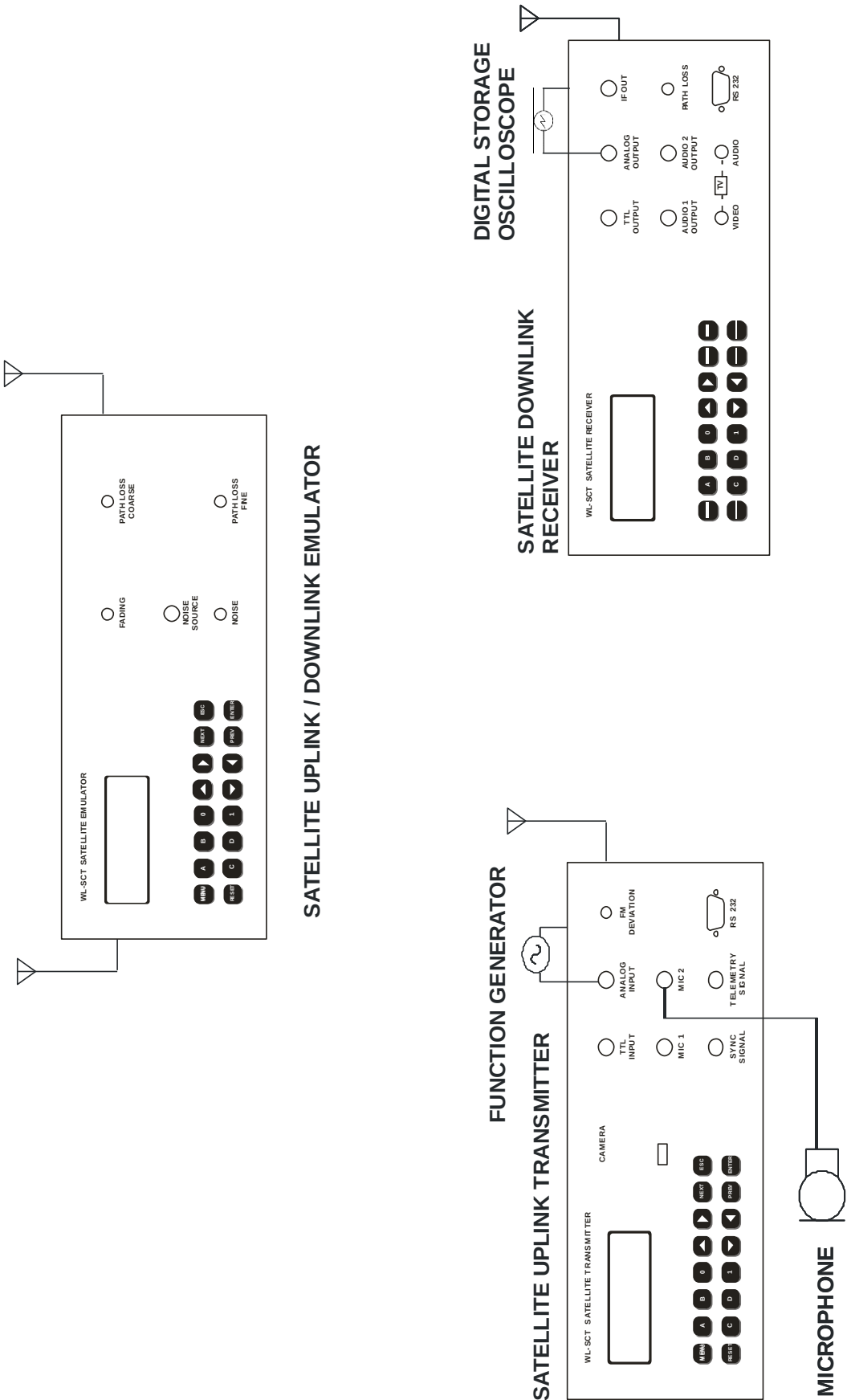
Down linking in commercial Ku band is at 11.700 – 12.200 GHz

In WL-SCT, up linking is carried out at 2.481 & 2.454 GHz whereas down linking is carried out at 2.400 & 2.427 GHz.

In WL-SCT the uplink and downlink frequencies are closer as compared to a commercial setup to conserve bandwidth and limit channel usage. The band pass filters inside the receiver and transmitter are real good with steep curves and accurate frequencies for optimum performance.

ISM (INDUSTRIAL, SCIENTIFIC & MEDICAL) band for satellite communication simulation is used as it is a license free band for institutional use. This band is from 2400 MHz to 2500 MHz.

EXPERIMENT NO. 2



Block diagram for Expt.2 Baseband analog signal

EXPERIMENT NO: 2

NAME

Baseband Analog Signal

OBJECTIVE

To measure the baseband analog signal parameters like Companding, Frequency Response of Audio and Video Channel, Cross Talk, Noise and Unclipped Sine wave in a satellite link.

EQUIPMENTS

- Satellite uplink transmitter, satellite downlink receiver and satellite link emulator with power cords.
- Antennas
- Microphone, 10 MHz Function Generator, Digital Oscilloscope.

PROCEDURE

- Setup the communication link as before.
- Set the Satellite Emulator Downlink frequency for 2.400 GHz.

	D	O	W	N	L	I	N	K		F	R	E	Q		B	A	N	D	
2	4	0	0		M	H	z		I	S		S	E	L	E	C	T	E	D

- Also make it sure that the PATH LOSS FINE and NOISE potentiometer is in fully anticlockwise position in order to receive better audio signals.
- Connect a microphone to MIC 2 in socket of Transmitter. Set the input channel for AUDIO CH2 for MIC 2.

				I	N	P	U	T		C	H	A	N	N	E	L	S		
A	U	D	I	O		C	H	1	:	O	F	F							
A	U	D	I	O		C	H	2	:	M	I	C	2						
V	I	D	E	O		C	H	3	:	O	F	F							

- Select Input channel for AUDIO CH2 to MIC 2 at Receiver end. Listen to the quality of voice spoken into the microphone at the TV speaker of the receiver in the AUDIO post. This establishes a **voice communication** satellite link.

				O	U	T	P	U	T		C	H	A	N	N	E	L	S	
A	U	D	I	O		C	H	1	:	O	F	F							
A	U	D	I	O		C	H	2	:	M	I	C	2						
V	I	D	E	O		C	H	3	:	O	F	F							

Steps for Companding

- Now connect 1 KHz 1V_{p-p} sine wave with a BNC-T connector to ANALOG INPUT of Transmitter so that the same sine wave signal can also be observed on one channel of DSO. Ensure that the level of sine wave fed is less than 1V p/p.
- Set the Input Channel in the Transmitter for AUDIO CH2 to ANALOG.

				I	N	P	U	T		C	H	A	N	N	E	L	S		
A	U	D	I	O		C	H	1	:	O	F	F							
A	U	D	I	O		C	H	2	:	A	N	A	L	O	G				
V	I	D	E	O		C	H	3	:	O	F	F							

- Make sure the potentiometer in the Emulator for the NOISE & PATH LOSS are in a fully anticlockwise position. Press A key from INPUT DATA menu for analog transmission through Emulator.

	I	N	P	U	T		D	A	T	A									
A	)		V	I	D	E	O												
B	)		T	E	L	E	M	E	T	R	Y		S	I	G	N	A	L	
C	)		V	I	E	W		R	X		T	E	L		D	A	T	A	

- Set the Input Channel in the Receiver for AUDIO CH2 to ANALOG.

				O	U	T	P	U	T		C	H	A	N	N	E	L	S	
A	U	D	I	O		C	H	1	:	O	F	F							
A	U	D	I	O		C	H	2	:	A	N	A	L	O	G				
V	I	D	E	O		C	H	3	:	O	F	F							

- Connect the ANALOG OUTPUT of Receiver to the other channel of CRO for comparing the transparency of signal received via satellite communication link. The transmitted & received waveforms are being monitored simultaneously on the same CRO on dual channel chopped mode.
- Measure the level of signal being transmitted and the level of signal being received.
- Find the levels of signals received on transmitting signal level of 50 mV to 2V in steps of 50mV. Draw a graph between transmitted and received signal levels. If the graph shows straight line then there is no companding being used in the system. Companding is used to increase the level of low level signals from microphones and reduce the level of high level signals in telephony.

Steps for observing Frequency Response of Audio CH2 Channels

- Apply 1 KHz, 1 V_{p-p} sine wave from function generator to ANALOG post of transmitter. Select CH2 to Analog in transmitter, Select VIDEO in Emulator and Select CH2 to Analog in receiver.
- Now measure the level of signal being received. Vary the frequency of input signal to 20 KHz in steps of 1 KHz and measure the level of received signals at different frequencies.
- Draw a graph between frequency of the input signal and level of the received signal. This would show the frequency response of the communication link for AUDIO CH2 channel.
- Measure the –3dB bandwidth of the audio channel. –3dB would be the level for which the measured signal is 30% lower than its level at a reference frequency of say 1 KHz.

- For AUDIO CH 1 channel analog frequency response is limited to 3.5 KHz because of the internal CODEC in Emulator.

Steps for Observing Frequency Response of Video Channel:

- Apply 10 KHz, 1 Vp-p sine wave from function generator to ANALOG post of transmitter. Select CH3 to Analog in transmitter, Select VIDEO in Emulator and Select CH3 to Analog in receiver.
- Adjust the received signal level to 1Vp-p by FM deviation pot at transmitter & Path Loss Course, Fading pot at Emulator. Measure the level of signal being received. Vary the frequency of input signal to 10 MHz in steps of 100 KHz and measure the level of received signals at different frequencies.
- Draw a graph between frequency of the input signal and level of the received signal. This would show the frequency response of the communication link for VIDEO CH3 channel.
- Measure the –3dB bandwidth of the video channel. –3dB would be the level for which the measured signal is 30% lower than its level at a reference frequency of say 10 KHz.
- Find the difference in frequency response from audio channels. Does the video channel show a better frequency response at higher frequencies? Does it correlate to difference in frequencies of audio and video signals?
- Compare the signal bandwidth at Receiver through wireless and through satellite link.

Steps for Observing Cross Talk

- Apply 1 KHz, 1 Vp-p sine wave from function generator to ANALOG post of transmitter. Select CH1 to Analog in transmitter, Select VIDEO in Emulator and Select CH1 to Analog in receiver.
- Measure the received signal in audio2 channel for different levels of FM deviation and mark this as V2.
Observe the received signal at Audio1 post of receiver and mark this as V1.
- Find the ratio of received signal in audio1 channel to the signal in audio2 channel.
 $\text{Crosstalk} = V1 / V2 = [20 \text{ Log}_{10} (V1 / V2)] \text{ dB}$
That would be a measure of cross-talk or channel separation.

Steps for observing Unclipped Sine wave

- Apply 1 KHz, 1 Vp-p sine wave from function generator to ANALOG post of transmitter. Select CH2 to Analog in transmitter, Select VIDEO in Emulator and Select CH2 to Analog in receiver.
- Observe the demodulated signal at the AUDIO2 OUTPUT post of the Receiver. Increase the Amplitude of the Sine Wave above 1Vpp.
- Observe the effect on the Amplitude level of the received signal at the Receiver. Find the maximum Amplitude of the Sine Wave above which the received signal starts getting clipped.

Steps for observing effect of Noise

- Apply 1 KHz, 1 Vp-p sine wave from function generator to ANALOG post of transmitter. Select CH2 to Analog in transmitter, Select VIDEO in Emulator and Select CH2 to Analog in receiver.
- Set up uplink and downlink as per expt.1.
- Note down the Noise Level at ANALOG OUTPUT of receiver in absence of input signal at transmitter.

- Apply 1 KHz, 1Vpp sine wave to ANALOG INPUT post of transmitter.
Set the Input Channel in the Transmitter for VIDEO CH3 to ANALOG.

				I	N	P	U	T		C	H	A	N	N	E	L	S		
A	U	D	I	O		C	H	1	:	O	F	F							
A	U	D	I	O		C	H	2	:	O	F	F							
V	I	D	E	O		C	H	3	:	A	N	A	L	O	G				

- Observe the demodulated signal with noise at the ANALOG OUT post of receiver.
- Switch off the emulator and tune Satellite Transmitter and Receiver for the same frequency (2.400GHz, 2.427GHz, 2.454GHz or 2.481GHz).
- Note down the noise level at ANALOG OUTPUT post of receiver in absence of input signal at transmitter.
- Apply 1 KHz 1Vpp sine wave to Analog input post of transmitter.
- Set the Input Channel in the Transmitter for VIDEO CH3 to ANALOG.

				I	N	P	U	T		C	H	A	N	N	E	L	S		
A	U	D	I	O		C	H	1	:	O	F	F							
A	U	D	I	O		C	H	2	:	O	F	F							
V	I	D	E	O		C	H	3	:	A	N	A	L	O	G				

- Observe the demodulated signal with noise at ANALOG OUTPUT post of receiver.
- Set up uplink and downlink as per expt.1.
- Set the Input Channel in the Transmitter for VIDEO CH3 to VIDEO.

				I	N	P	U	T		C	H	A	N	N	E	L	S		
A	U	D	I	O		C	H	1	:	O	F	F							
A	U	D	I	O		C	H	2	:	O	F	F							
V	I	D	E	O		C	H	3	:	V	I	D	E	O					

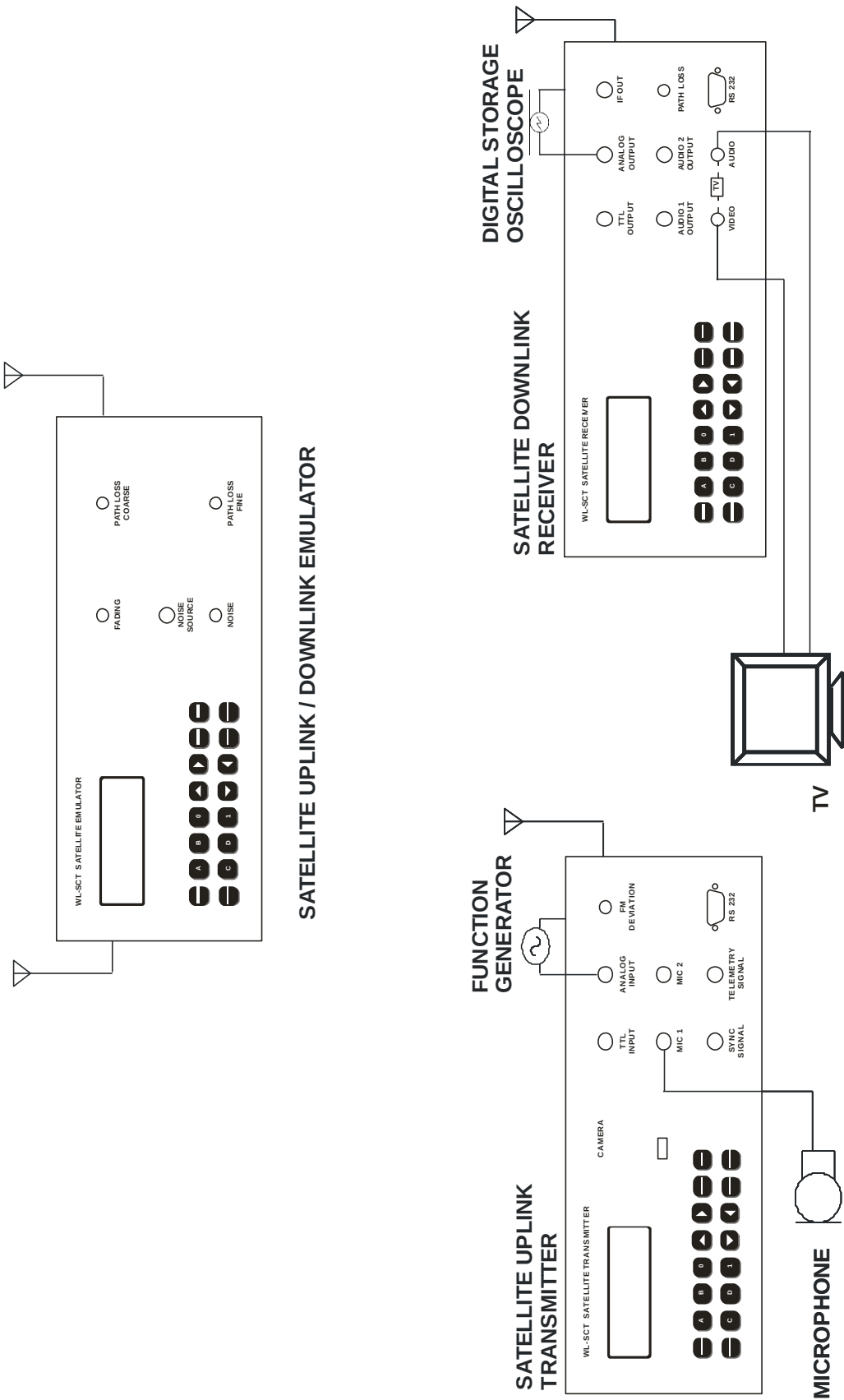
- Observe the demodulated signal on the screen of the DSO at receiver.
- Introduce noise by rotating NOISE potentiometer clockwise and observe the effect on picture as noise goes on increasing.

RESULT

The sound into the microphone is converted into electrical signal and FM modulated onto a sub-carrier of 6 or 6.5 MHz. The sub-carrier is then mixed with main carrier at 2.454 or 2.481 GHz. The main carrier is carrying the video signal on FM modulation. The same holds true for any other audio signal also. The mixing of sub carrier generates signals of say 2.4475, 2.448, 2.454, 2.460 & 2.4605 GHz with main carrier at 2.454 GHz. The sub-carriers are at a level of around 20-25dB lower than the main carrier. For this reason audio channels are more prone to fading. The modulated carrier is then radiated from the antenna and received by the satellite transponder. The satellite then transverts this carrier to another frequency and retransmits the amplified signal to receiving base station at different frequency. This frequency is then received by the earth station and demodulated to give the audio output. Baseband analog (Voice) Signal could be received only because the Transmitter, Uplink satellite link emulator, Downlink satellite link emulator and Receiver all are PLL locked to accuracy of less than 10 KHz.

The system can transmit and receive two audio signals and a video signal simultaneously. The nominal level it can support is 1V for audio & video channels. It can support an audio signal well up to 20 KHz and a video signal well up to 5MHz. Both the audio channels are almost identical. Signal quality degrades slightly when passing through satellite. Cross-talk in audio channels is around 30dB.

EXPERIMENT NO. 3



Block diagram for Expt.3 Analog FM/FDM TV Satellite link

EXPERIMENT NO: 3

NAME

Analog FM/FDM TV Satellite Link

OBJECTIVE

To measure the signal parameters in an analog FM/FDM TV Satellite link and to study the functionality of a satellite MODEM

EQUIPMENTS

- Satellite uplink transmitter, satellite downlink receiver and satellite link emulator
- Antennas
- Microphone, video monitor, Function generator, Digital oscilloscope X 2, spectrum analyzer

THEORY

Bandwidth

The bandwidth of a microwave signal is relatively large compared with its terrestrial AM counterpart, and is normally in the range 24-36 MHz. For medium power FSS and DBS satellites, a transponder of around 27 MHz is commonly used, although a few (the Eutelsat II series for example) have bandwidths of 36 and some of 72 MHz. With 72 MHz channels it is possible to transmit two 36 MHz bandwidth channels using the same transponder (so-called half-transponder format). Since the frequency spectrum of a FM signal is infinite (produces an infinite range of sideband frequency components) an infinite bandwidth would be needed to transmit it. Clearly some form of compromise or band-limiting is necessary in practice, which must be related to the deviation value used. From subjective tests, it has been found that picture quality derived from 27 MHz channels is indistinguishable from that of 36 MHz or more, and that bandwidths as low as 16 MHz still produce reasonable picture quality. In fact some receivers allow the user to reduce the bandwidth of the IF filter to 15 or 16 MHz to reduce noise, thus increasing the pre-detection C/N ratio. The trade-off with wide bandwidths is a correspondingly lower number of channels that may be fitted into a given frequency allocation. Higher 36 MHz bandwidth signals produce a better improvement in the S/N ratio on demodulation than do 27 MHz signals (FM improvement), so a particular value of S/N can be achieved with a lower C/N ratio.

Deviation

With frequency modulation, the instantaneous frequency of the carrier signal is varied in response to the instantaneous voltage of the video signal (including sync tips). This modulation method produces an infinite number of frequency components as sidebands. The amplitude of these components decreases with the distance from the carrier frequency. For practical purposes, only a limited number of these components need be sent without affecting the perceived picture quality. Band-limiting these smaller components produce

very little distortion and a minimum bandwidth, somewhat larger than the maximum deviation, is normally sufficient. The maximum frequency deviation of the modulated signal is the frequency difference between the maximum modulated frequency and the unmodulated frequency and corresponds to the maximum and minimum amplitudes of the message signal, respectively. The ratio of peak deviation and the highest video frequency is called the frequency modulation index. This depends on the sensitivity of the modulator, and increasing this has the effect of spreading out the signal spectrum. Increasing the deviation of the transmitted signal results in a higher S/N ratio (less noise). FM deviation is a measure of the modulator sensitivity (in units of MHz/V but is often quoted in MHz). This assumes that the peak-to-peak value of the video signal is 1 V, including synchronization pulses.

In a link budget calculation, we need the peak-to-peak deviation value, $\Delta f(p-p)$, of the video signal (in Hz) in order to calculate the S/N ratio after demodulation in the receiver. If a peak deviation value is quoted, remember to double it to obtain the peak-to-peak value (sync tips to peak white).

With satellites operating on the half-transponder format the FM deviation value may be reduced {halved} to simulate the effect of reduced S/N since signals from two channels are modulated onto the same carrier. The half-transponder format is where two channels are simultaneously modulated onto a single, say, 72 MHz bandwidth transponder.

Estimating FM deviation

If the FM deviation, or video deviation is not known, but you know the bandwidth of a required channel you can use Carson's rule to arrive at a reasonable estimate of the peak-to-peak frequency deviation. FM deviation is defined above.

Deviation (peak-to-peak) = RF bandwidth - 2 (maximum video frequency) (Hz)

Example 1

Astra 1a (Europe) uses 26 MHz bandwidth channels for 5 MHz video:

Video deviation (peak-to-peak) = $26 - 2(5) = (26 - 10) = 16$ MHz

The quoted figure is 16 MHz/V (standardized video signals are typically 1Vp-p amplitude including sync pulses).

Example 2

The Eutelsat H series uses 36 MHz bandwidth transponders for 5 MHz video:

Video deviation (peak-to-peak) = $36 - 2(5) = (36 - 10) = 26$ MHz The quoted figure is 25 MHz/V so, as you can see, a reasonable approximation is obtained by using Carson's rule.

PROCEDURE

- To set the Video Link, set the Transmitter & Emulator Uplink Frequency to 2481 MHz, and Receiver & Emulator Downlink frequency to 2400 MHz. This is done to ensure the emulator downlink PLL is locked and displayed frequency is generated correctly.
- If you get the picture on the TV screen at the receiver via satellite, PLL of complete link are O.K. and a successful satellite link is said to be established.
- Set the input channel for VIDEO CH3 of the Transmitter to video.

				I	N	P	U	T		C	H	A	N	N	E	L	S		
A	U	D	I	O		C	H	1	:	O	F	F							
A	U	D	I	O		C	H	2	:	O	F	F							

V	I	D	E	O		C	H	3	:	V	I	D	E	O					
---	---	---	---	---	--	---	---	---	---	---	---	---	---	---	--	--	--	--	--

- Feed a 2 KHz 1Vp-p sine wave externally at ANALOG INPUT of the Transmitter, set AUDIO CH2 to ANALOG and set the input channel for AUDIO CH1 to MIC 1.

				I	N	P	U	T		C	H	A	N	N	E	L	S		
A	U	D	I	O		C	H	1	:	M	I	C	1						
A	U	D	I	O		C	H	2	:	A	N	A	L	O	G				
V	I	D	E	O		C	H	3	:	V	I	D	E	O					

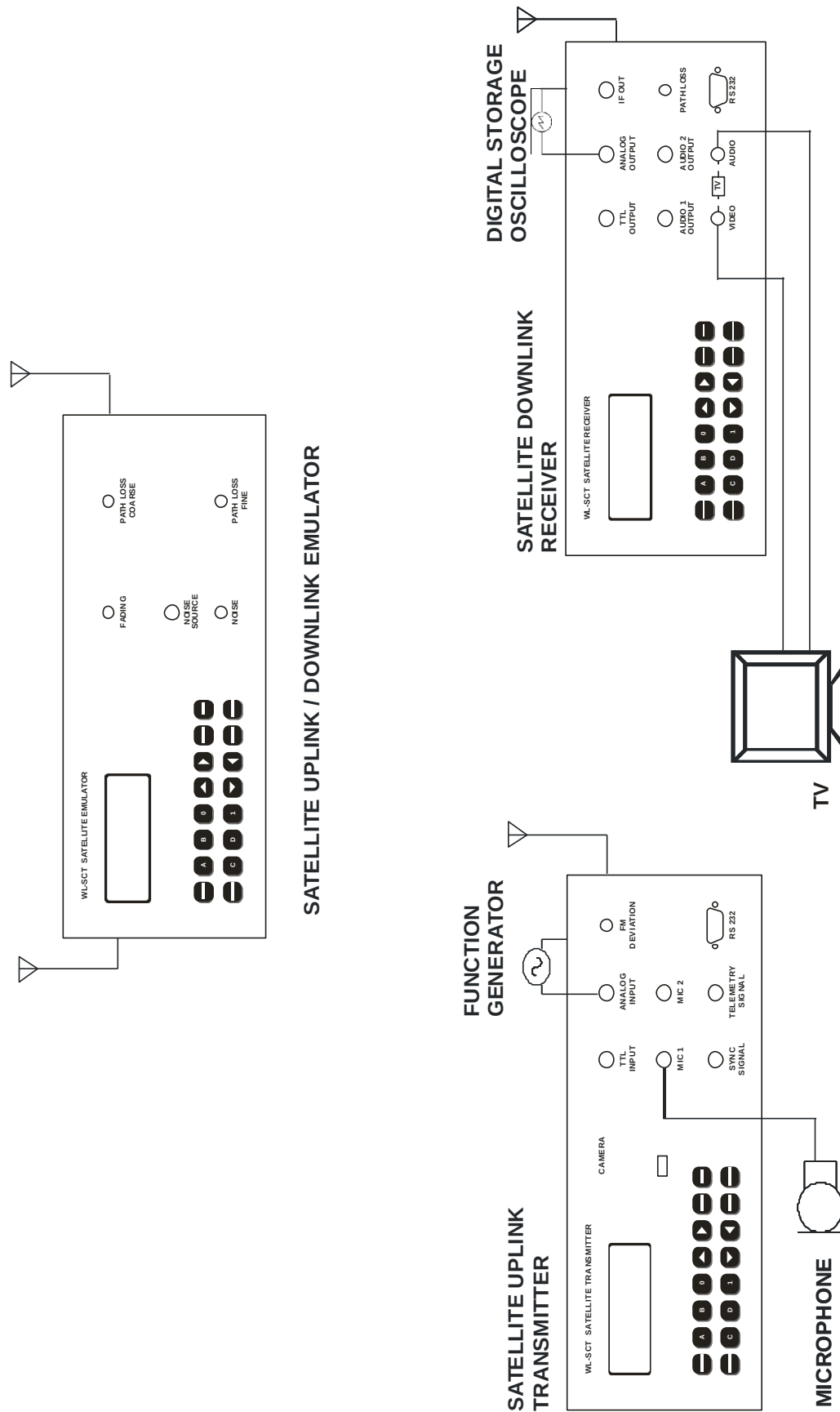
- Connect the video Monitor to video out of Receiver and connect the power supply of Monitor.
- See if you are able to receive both the audio & video sent at different channels clearly. This would perform the functionality of a satellite MODEM: Modulating the base band on a carrier at Transmitter end and Demodulating the received carrier at Receiver end after being passed through satellite.
- See if you can receive video as well as both audio frequencies simultaneously. This is a complete Analog FM / FDM TV satellite communication link. In commercial broadcast the two audio channels are the left & right stereo channels and the video is the motion picture, which together comprise the signal content.
- Connect DSO to SYNC SIGNAL of transmitter and Find the sync. Level of video signal fed. If you put a black sheet of paper or your hand in front of CCD camera so that no light can enter into lens of camera then negligible signal is present to modulate the video carrier. Therefore, what you see on display of DSO is the internal sync. Level of camera. Measure how much mV is it.
- If you vary intensity of light in front of camera, meaning that you remove black sheet in front of lens. Then varying light from object etc. will modulate the video signal and you will see a continuously varying complex video signal on DSO retrieved at Receiver end. Bringing your hand in front of camera and taking it away will vary the FM deviation of signal.
- Observe if increasing or decreasing the video FM deviation from pot at Transmitter end any effect on parameters of has received sine wave.
- Observe on CRO, how video, audio/sine wave behaves on fading the carrier by introducing the Fading from satellite link emulator.

RESULT

This is an analog FM/FDM system where audio and video both are FM modulated on carrier at transmitter and relayed to satellite which then transponds the signal and sends it back to the receiving station. The system uses a channel allocation of 27 MHz as specified for satellite video link. Within this band there are audio sub carriers of 6 & 6.5 MHz, which can carry different audio channels simultaneously for different languages or stereo. FDM is implemented because three different frequencies are used for transmission of three separate signals. The video amplifier has a bandwidth of 5 MHz. The fm deviation is 4MHz for a video signal of 1V p/p.

The process of modulation and demodulation is analog FM with wide bandwidth for video signal and narrow bandwidth for audio signal. The FM demodulation is carried out using PLL demodulators for wide band response and good linearity.

EXPERIMENT NO. 4



Block diagram for Expt.4 Carrier to noise ratio

EXPERIMENT NO: 4

NAME

Carrier To Noise Ratio

OBJECTIVE

To measure the C/N ratio

EQUIPMENTS

- Satellite uplink transmitter, satellite downlink receiver and satellite link emulator
- Antennas
- Microphone, video monitor, Function generator, oscilloscope X 2, spectrum analyzer

THEORY

Carrier-To-Noise Ratio

For the Ku and Ka bands the system carrier-to-noise (C/N) ratio is given by:

$$C/N = EIRP - L_{fr} + G/T_{usable} - 10 \log(kB) - A_{rain} - A_{atm} \text{ (dB)}$$

where : EIRP = the equivalent isotropic radiated power from the satellite at the site location (dBW)

L_{fr} = free space path loss on the earth to satellite path (dB)

G/T_{usable} = minimum degraded value of the system figure of merit (dB/K)

k = Boltzmann's constant (1.38×10^{-23} J/K)

B = receiver's pre-detection intermediate frequency (IF) bandwidth (Hz)

A_{atm} = gaseous attenuation due to atmospheric absorption (dB)

A_{rain} = rain attenuation for a given percentage of the time (dB).

Note: (a) A_{rain} & A_{atm} can be omitted for operation frequencies of <8 GHz; and

(b) for a 'clear-sky' calculation omit the A_{rain} term and substitute the nominal figure of merit, $G/T_{(nominal)}$, for $G/T_{(usable)}$.

Antenna Noise

Any signal received is combined with an element of noise, which degrades the overall performance:

Signal = wanted signal + noise

Obviously, the noise component must be kept as small as possible, taking into account cost and available technology. Noise can come from many sources and is produced by the thermal agitation of atoms and molecules above absolute zero (-273°C or 0 K ; note that the degree sign is not used on the Kelvin scale). This is why noise is said to have an equivalent noise temperature. The noise temperature of the earth is normally standardized at 290 K (17°C). There are three main sources of noise in the environment:

1. Extraterrestrial noise sources:- This is wide bandwidth radiation caused by the energy conversion in stars and the residual back-ground radiation of the 'big bang'. This tends to taper off at 1 GHz and settles to that of the residual background noise alone which is taken as 2.7 K . Above 2 GHz , there are only a few isolated points of very strong non-thermal

noise, principally from Cygnus A, Cygnus X, Cassiopeia A and the Crab nebula. There is also a narrow band of increased noise from the Milky Way. The Sun is an enormous source of noise at around 10,000 K at 12GHz and the Moon at about 200 K. **This noise enters the antenna mainly via the main lobe.**

2. Man-made noise:- This noise emanates from microwave pollution due to man's electrical activities and principally **enters the antenna via the side lobes.**

3. Ground noise:- In the long term, this is the major component of noise incident on the antenna aperture, and depends mainly on the antenna diameter, antenna depth, and elevation setting. The smaller the diameter of the dish the wider and more spread out will be the side lobes, so more noise will enter from the warm earth. The noise temperature also increases as the elevation angle decreases, since lower elevation settings will pick up more ground noise due to side lobes intercepting the ground (diffraction effects at the antenna rim). This may be reduced by various methods of feed illumination. The design of the antenna itself also plays a part. A deep dish picks up less ground noise at lower elevations than do shallow ones, also prime focus mounted head units will add to noise since it is 'seen' at the same temperature as the Earth. Inclining the head unit away from the earth and towards the cool sky as happens in the case of an offset focus design can also improve things. This practice tends to counteract the negative effects of increased beam width for small antennas set at low elevation angles.

Noise And Its Effects

Anybody, above the temperature of 0 K or -273°C has an inherent noise temperature. Only at absolute zero temperature does all molecular movement or agitation cease. At higher temperatures molecular activity causes the release of wave packets at a wide range of frequencies some of which will be within the required bandwidth for satellite reception. The warmer the body the higher the equivalent noise temperature it will have, resulting in an increase in noise density over the entire spectrum of frequencies. The warm earth has quite a high noise temperature of about 290 K and consequently rain, originated from earth, has a similar value. **The characteristic appearance of noise on FM video pictures can be either black or bright white tear drop or comet shaped blobs (sparkles) that appear at random on the screen.** It is subjectively far more annoying than the corresponding snowy appearance of noise on terrestrial AM TV pictures. Video cassette recorder pictures, also frequency modulated, display annoying sparkles as a result of worn/dirty heads or faulty head amplifiers. Only relatively small amounts of FM noise can be tolerated.

Free Space Path Loss

As the radiated signal of a transponder travels towards earth it loses power by spreading over an increasingly wide area thus diluting the signal strength. This effect is known as the free space path loss and the greater the distance the receiving site from the satellite the more it increases. Contributory factors include absorption of microwaves by gases and moisture in the atmosphere. The power density of signals, measured in watts per square meter, finally arriving at earth is extremely weak.

Rain Attenuation

One of the major problems with satellite reception is rain, and to a lesser extent snow and hail. The weak incoming microwave signals are absorbed by rain and moisture, and severe rainstorms occurring in thunder conditions can reduce signals by as much as 10 dB (reduction by a factor of 10). Not many installations can cope with this order of signal reduction and the picture may be momentarily lost. Even quite moderate rainfall can reduce signals by 2 to 3 dB which is enough to give noisy reception on some receivers. Another

problem associated with rain is an increase in noise due to its inherent noise temperature, which is similar to that of the earth. In heavy rain depolarization of the signal can also occur resulting in interference from signals of the opposite polarization but same frequency. This effect is more noticeable with circular polarization.

Factors affecting satellite reception:-

The performance of a satellite TV receives only (TVRO) system is affected by a number of physical factors. Some of these are outlined below:

- The equivalent isotropic radiated power (EIRP) of the satellite.
- The effective antenna diameter.
- The low noise block (LNB) noise figure or noise temperature.
- Coupling losses by waveguides and Polarizers.
- Antenna pointing losses: initial pointing error (degrees).
- Antenna stability in wind or other environmental conditions (degrees).
- Satellite station keeping accuracy.
- Polarization losses.
- Transponder ageing.
- Rain attenuation for signal availability (typically 99.5% of average year).
- For Ku and Ka band, noise increase due to precipitation (rain, snow or hail).
- Atmospheric absorption by oxygen and water vapor (depends on humidity).
- Temperature variations.
- The receiver (demodulator threshold) figure.
- The signal modulation characteristics.
- Scattering of signals due to blockages such as trees, buildings, birds and aircraft.
- Spreading loss through the atmosphere.

Transient effects such as passing birds and aircraft are largely unpredictable so can be neglected from the calculation. The others can all have a significant long-term effect, although factors 8, 9 and 10 can be neglected for S and C band reception.

Downlink Path Distance

The path distance, sometimes called the 'slant range', is the distance between the ground station and the satellite of interest. Clearly the further away from the equator this is, the longer the path distance. An equation used to calculate this is:

$$\text{Path distance (D)} = 6378.16 \sqrt{(m^2 + 1 - 2M [\cos(A) \cos(B)])} \text{ (km)}$$

Wavelength:-

In many equations, including those that follow, a wavelength (λ) value rather than frequency is required for simplification. Conversion from frequency to wavelength can be done using:

$$\lambda = C/F$$

Where: c = the speed of light (2.998×10^8 m/s), F = frequency (Hz).

Free Space Loss

The free space loss (L_{FS}), or path loss, expresses the attenuation of microwave signals on their Earth-bound journey and occurs due to the spreading out of the beam. A good analogy is visualized by the intensity fall-off of a car headlight beam with distance. The path loss increases with frequency and is greatest for low antenna elevation angles. A suitable equation for calculating its value is:

$$L_{FS} = 20 \log [(4000\pi D)/\lambda] \text{ (dB)}$$

Where: $\pi = 3.14159$

D = path distance (km)

λ = wavelength (m).

Antenna Gain

The antenna gain (Ga) increases with the effective antenna size which takes into account the efficiency (p) of the antenna. The gain can be expressed as:

$$\text{Antenna gain (Ga)} = 10 \log \{(\pi \cdot d)^2 p / 100 \lambda^2\} \text{dB}$$

Where d = the antenna diameter (m)

p = the percentage antenna efficiency (60-80% typically)

λ = wavelength (m)

Note: the antenna efficiency may be specified as a normalized value less than 1 (e.g. 0.67 or 0.80) rather than as a percentage. In such cases delete the term 100 in the denominator and substitute the normalized factor for p.

Effective Antenna Noise Temperature

The effective antenna noise temperature (Ta) defined above is now discussed in a little more detail. The effective antenna noise temperature is determined by many factors, such as antenna size, elevation angle, external noise sources and atmospheric propagation effects. During clear-sky conditions, the principal noise component of the effective antenna noise temperature is ground noise pick-up. This is easy to see since, neglecting atmospheric propagation effects (rain, etc), this is virtually all the noise entering the antenna. This is the 'antenna noise' parameter that manufacturers often tabulate for a range of elevation angles; it may also include a relatively small contribution by galactic background noise. There are three main contributions to the overall antenna noise :

1. Antenna noise temperature due to ground noise (Tant): The smaller the antenna, the wider and more spread out is the side lobes intersecting the warm earth, and, consequently, the more ground noise is picked up by the antenna. It can also be seen that these side lobes, principally the first side lobe, would intersect the ground at a higher elevation angle than that of a larger antenna and so would be a noisier device when set at a given elevation. Ground noise pick-up may be reduced, at the expense of gain, by under-illuminating the dish; thus, this factor essentially determines the efficiency of the dish. Size being equal, a prime focus antenna would detect increased ground noise over an offset design since the head unit, directly mounted in the signal path, would be 'seen' at the same temperature as the Earth.

Since the antenna noise temperature has so many variable factors, it is apparent that in the absence of a manufacturer-supplied figure, an estimate is perhaps the best we can hope for. Equation takes into account the elevation and the diameter, may be used to calculate a reasonable approximation for the antenna noise under clear-sky conditions.

$$T_{ant} = 15 + 30/D + 180/EL(K)$$

where: D = antenna diameter (m)

EL = dish elevation angle (degrees)

2. Cosmic or galactic noise component: This is background cosmic noise, principally the residual noise of the 'big bang'. It has a small noise temperature of about 2.7 K. This component is relatively small in relation to the error in estimating the ground noise component, and may be omitted from practical calculations. In any case, depending on how 'antenna noise' is defined in manufacturers' specifications, this may be incorporated.

3. Atmospheric propagation components: There are two main propagation effects experienced on the downlink. Firstly, atmospheric gaseous absorption by water vapor and oxygen; this is basically a clear-sky effect. Its value depends on the absolute humidity or vapor density measured in grams per square meter, the antenna elevation and the frequency involved. It is a relatively minor contributor below about 7.5 GHz. The second propagation effect is attenuation due to precipitation. Considering the uplink situation, a

receiver on board a satellite will 'see' a fairly constant but high noise temperature emitted from the warm Earth of around 290 K, so further thermal energy emission by rain will have a negligible effect. In the downlink situation, the receiver is directed toward a relatively cool sky so, in a relative sense, the additional thermal noise contribution by rain is by no means a negligible component of the total system noise, especially if the receiver (LNB) is a low noise device operating in the Ku or Ka band. **The effects of rain and atmospheric absorption are negligible in the S and C bands.**

Precipitation will not only directly attenuate the signal (known as a 'rain fade'), but the system noise temperature will also increase since the temperature of the intervening medium approaches that of the Earth. It is important that the increase in system noise is taken into account and not just the attenuation experienced by a rain fade. The combination of the two is known as the **"downlink degradation (DND)"**

The effects of precipitation become significant above about 8 GHz. Rain, or to a lesser extent snow, fog, or cloud, attenuate and scatter microwave signals. The magnitude depends more on the size of the water droplets (in cubic wavelengths) rather than the precipitation rate itself. Heavier rain tends to comprise larger droplets so the two are normally related. As a general rule, the physical-medium temperature, of all forms of precipitation, is taken as 260 K. For clouds and clear-sky use 280 K.

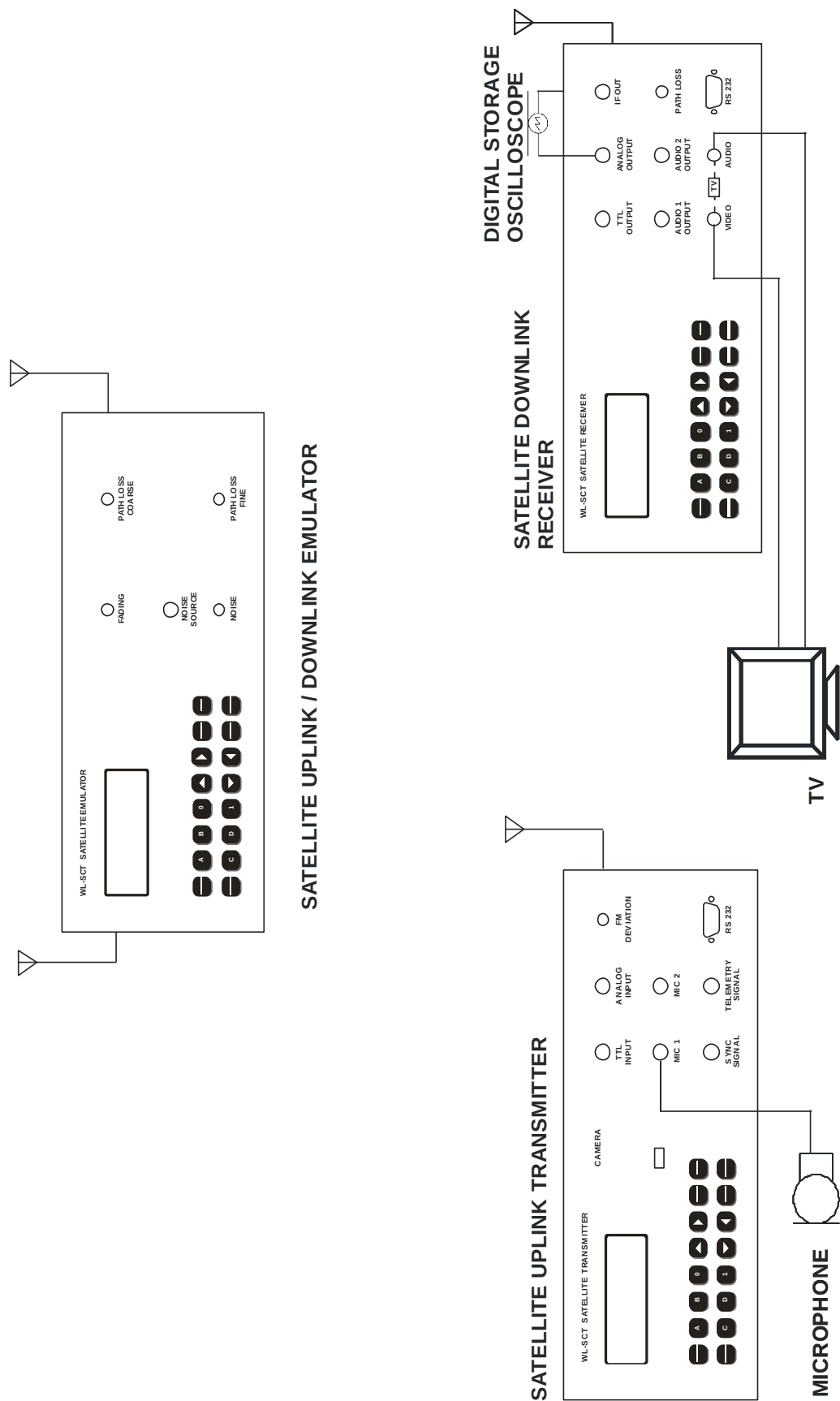
PROCEDURE

- To set the Video Link, set the Transmitter & Emulator Uplink Frequency to 2481 MHz, and Receiver & Emulator Downlink frequency to 2400 MHz. This is done to ensure the emulator downlink PLL is locked and displayed frequency is generated correctly. If you get the picture on the TV screen at the receiver via satellite, PLL of complete link are O.K. and a successful satellite link is said to be established.
- Now, switch off the carrier by switching off both Transmitter and satellite Emulator. Receiver will read only its noise floor on RSSI menu.
To view the RSSI menu, press 'C' Key from main menu then press 'B' key in Receiver. Say, in absence of any carrier, Receiver reads 0.92 V which corresponds to -96 dBm. Thus, -96 dBm is noise floor of Receiver that means if carrier received by Receiver is less than -96 dBm it will be unable to measure it.
- Now, switch ON Transmitter and satellite Emulator and say, the Receiver reads -59 dBm (1.93 V) of carrier level being received. Thus, $C/N = \text{carrier level} / \text{noise level}$. As both noise and carrier signal detected are measured in dB, C/N can be calculated by taking the difference of two readings or $C/N = \text{carrier level (in dB)} - \text{noise level (in dB)}$. Hence, $C/N = -59 - (-96) = 37 \text{ dB}$.
- Make sure the Receiver is not saturated with carrier otherwise incorrect C/N will be read. This can be done by increasing path loss at satellite Emulator and or taking Receiver farther away from satellite Emulator.
- Measure the C/N readings for different levels of path loss.
- Monitor the audio and video transmissions and correlate them to various levels of C/N. Does higher level of C/N result in better picture and sound quality?
if you are able to receive audio & video sent, clearly it means you are well above threshold level of signal. Now, the effect of noise can be seen if you decrease the received signal strength to a considerable level. This can be achieved by increasing the path loss.
This means the received signal is just above the noise floor of receiver. Although we are using FM demodulator but because the received signal is barely above the noise floor you can hardly receive any intelligent information. Thus, signal cannot be received below noise floor of Receiver.

RESULT

The difference between two readings of receiver noise level and carrier level is the C/N ratio in dB. Actual reading will depend on a number of factors and will differ from case to case. Increasing the path loss and distance between antennas shall result in lower C/N ratios due to lower levels of received carrier. Amount of noise received/generated remains constant. More power at transmitter shall result in better picture quality and more C/N ratio. Lower noise at receiver is essential for better picture. Higher gain antenna could be used to capture more signal. Hence a helix antenna could result in higher C/N. Sparkles start appearing on black or white portions of picture when noise is increased. Further increasing the noise will make the picture lose its sync resulting in complete loss of information.

EXPERIMENT NO. 5



Block diagram for Expt.5 Signal to Noise ratio

EXPERIMENT NO: 5

NAME

Signal to Noise Ratio

OBJECTIVE

To measure the S/N ratio

EQUIPMENTS

- Satellite uplink transmitter, satellite downlink receiver and satellite link emulator
- Antennas
- Microphone, video monitor, Function generator, oscilloscope X 2, spectrum analyzer

THEORY

Signal to Noise Ratio (S/N Ratio)

This is the ratio of the desired signal E.M.F. to any noise E.M.F. present. It should be as high as possible. If this ratio falls to unity or below, the signal is rendered virtually useless. (It is possible, but expensive, to use computer generated 'signal enhancement' techniques in some cases, but for domestic satellite broadcasting this is out of the question).

Providing the individual deviations of a small number of audio channels are small in relation to the video deviation, it is assumed for practical purposes that the overall peak-to-peak deviation of a baseband signal (including the multiple sound carriers) approximates that of the video signal alone.

For frequency modulated (FM) television signals, the signal-to-noise (S/N) ratio on demodulation can be calculated as:

$$S/N = C/N + 10 \log [3\{f(p-p)/f_v\}^2] + 10 \log (b/2f_v) + K_w \text{ (dB)}$$

where: S/N = the peak-to-peak luminance amplitude to weighted r.m.s. noise ratio (dB)

C/N = carrier-to-noise ratio (dB)

f(p-p) = peak-to-peak deviation by the video signal including the sync pulses (Hz)

f_v = highest video frequency present (Hz)

b = radio frequency bandwidth (usually taken as f(p-p) + 2f_v (Hz)

k_w = combined de-emphasis and weighting improvement factor in FM systems (dB).

Note: (a) above Equation is only valid for systems operating above the demodulator threshold. (b) The effect of the additional deviation for multiple sound sub-carriers located above the video baseband tends to improve the video S/N ratio slightly (by a fraction of a decibel) over that calculated using above equation. For practical purposes the overall peak-to-peak deviation may be taken as the overall peak-to-peak deviation by the video signal, provided the individual deviations of the audio channels is small in comparison; (c) The combination of the second and third terms of Equation is sometimes called the 'FM modulation gain' or 'FM improvement'.

Signal Availability and Operational Margins

An attenuation figure for rain has to be predicted from long-term rainfall statistics for the receive site of interest. Rather than allow a massive operational margin over threshold for the worst ever rain storm likely, we are normally content with specifying a signal availability figure for an average year, which potential customers find acceptable. In other words for a percentage of time the signal will not fall below some predetermined C/N (or S/N) ratio. For example, when we say a CCIR grade 4 (good) signal is available for 99.7% of an average year we mean that the S/N ratio is not expected to fall below 42.3 dB for 99.7% of the time (or 99% of the worst month). However, it will be expected to occasionally fall below this for 0.3% of the time during severe storms. The higher the signal availability designed into a system, the better will be the protection against the effects of rain attenuation. The dish size needed also grows alarmingly as the designed signal availability increases. Rain attenuation, or more specifically the downlink degradation, is the major component of the overall loss margin for Ku and Ka band systems. For typical direct-to-home (DTH) systems, a figure of 99.5% availability is normally considered acceptable. In fact most packaged fixed dish systems for popular satellites are designed around this figure. For satellite master antenna TV (SMATV) you may require a higher figure of 99.9%, and for cable head even higher. The law of diminishing returns eventually intervenes since 100% availability is impractical.

Noise Weighting Factor

When a high bandwidth signal is transformed to a lower baseband value, an increase in the S/N ratio is to be expected. Although the FM improvement value (also called FM modulation gain) may be calculated, viewers vary in their perception of differing spectra noise accompanying the video signal. As a result of many subjective tests, standardized noise weighting figures have been introduced for various TV systems to correct for this effect. Values are typically around 11.2 dB for PAL I, 10.2 dB for NTSC M, and 13 dB for MAC.

C/N, S/N and Threshold

The carrier-to-noise ratio (C/N) is relevant before demodulation in the receiver. The signal-to-noise ratio (S/N) is that relevant after demodulation. The S/N ratio is thus dependent on both the C/N ratio and the modulation characteristics.

Another important link parameter is the receiver's demodulator 'threshold' figure. Threshold is the point where the linear relationship between demodulator C/N input and S/N output begin to break down. The demodulator threshold is the point at which the demodulator in the receiver loses its linear relationship between input C/N and output S/N. Thus if a system is operating near or below threshold a small temporary reduction in C/N caused by rain, etc., can result in a non-linear reduction in S/N. If the C/N sinks below threshold then the calculated S/N value is invalid. At the time of writing, typical values obtained using extended threshold demodulators are in the range 5-6 dB.

Nominal Figure of Merit

G/T is the ratio of the net antenna gain and total system noise temperature. The 'nominal figure of merit' (G/T nom.) is the maximum obtainable figure for a given elevation angle and comprises the net antenna gain (antenna gain-coupling loss) divided by a noise temperature factor made up from contributions of the equivalent receiver noise temperature (i.e. LNB), the coupling noise of inserted Polarizers and waveguide components such as orthomodal transducers (OMTs) and the 'clear sky' modified antenna noise temperature. No operational margins are included such as antenna misalignment losses, ageing, or the increase in antenna noise for a given percentage of time due to rain. This is the highest value of the G/T ratio allowing qualitative comparison between different outdoor units. The higher the ratio, the better the system will perform. G/T, in general, is the figure, which has the greatest effect on the final C/N ratio. All other contributory factors are relatively constant.

$$G/T \text{ nom} = 10 \log [10^{0.1(G+a)} / T_{\text{sys}}] (\text{dB/K})$$

Where G = antenna Gain (dB)

a = coupling loss (dB) by waveguide components (loss = negative gain)

T_{sys} = clear sky system noise temp. Excluding propagation effects

Usable Figure Of Merit

The Required G/T Parameter needed in a detailed link budget is the 'usable (degraded or minimum) figure of merit' (G/T usable) this allows for further operational losses due to antenna pointing errors, polarization effects, ageing, and the increase in system noise due to precipitation for a given percentage of time and comprises the net antenna gain (antenna gain -coupling loss -operational losses) divided by the total system noise temperature. This G/T thus characterizes the 'in service' performance and is the one used in detailed link budgets. An additional noise temperature contribution is added to T_{sys} to allow for the increase in system noise due to precipitation for a certain specified percentage of the time. This is expressed mathematically by:

$$G/T_{\text{usable}} = 10 \log [10^{0.1(G+a+b)} / T_{\text{sysrain}}] (\text{dB/K})$$

Where G = antenna Gain (dB)

a = coupling loss (dB) by waveguide components (loss = negative gain)

b = losses due to antenna pointing errors, polarization errors and ageing (dB) (loss = negative gain)

T_{sysrain} = modified total system noise temperature which includes the increase in noise temperature due to precipitation for a given percentage of the time (K).

PROCEDURE

- To set the Video Link, set the Transmitter & Emulator Uplink Frequency to 2481 MHz, and Receiver & Emulator Downlink frequency to 2400 MHz. This is done to ensure the emulator downlink PLL is locked and displayed frequency is generated correctly.
- If you get the picture on the TV screen at the receiver via satellite, PLL of complete link are O.K. and a successful satellite link is said to be established.
- Remove cables from TTL INPUT, ANALOG INPUT, MIC 1 and MIC 2. Remove Video signal from channel 3 by making VIDEO CH 3 OFF.
- Measure the noise floor of all the base band outputs of Demodulator of Receiver by removing all modulating inputs at Transmitter and satellite link emulator, with the help of DSO. The DSO can measure the noise floors of each base band outputs in mV.
- Now, set the input channel for VIDEO CH3 of the Transmitter to ANALOG so that you will start receiving the modulated carrier.

				I	N	P	U	T		C	H	A	N	N	E	L	S		
A	U	D	I	O		C	H	1	:		O	F	F						
A	U	D	I	O		C	H	1	:		O	F	F						
V	I	D	E	O		C	H	3	:		A	N	A	L	O	G			

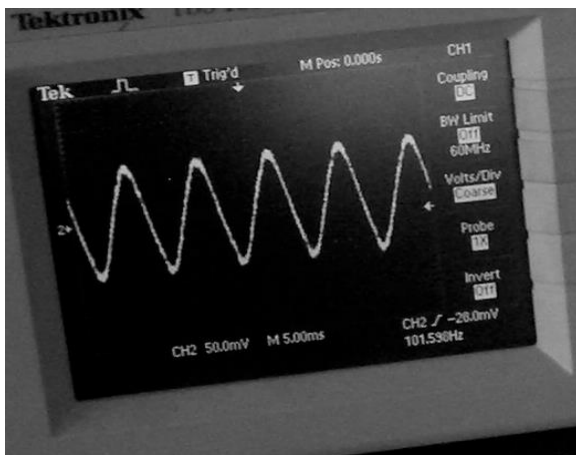
- The ANALOG out of Receiver will demodulate the received signal and extract the modulating signal. Analog signal can be measured using DSO.
- As both noise and modulating signal are measured in mV, actual signal (S) can be calculated by taking the difference of the two readings. Say, noise floor is 50mV and analog signal or sine wave at Rx is, as read on DSO, say, 1050 mV. Now, S is equal

to 1000mV. Now, S/N is 20 (I e ratio of signal to noise = 1050/50) and S/N in dB = $10 \log S/N$ in numerals. That is $20 \log 20 = 26\text{dB}$.

- Measure S/N by varying path loss in the Emulator.
- Monitor the audio and video transmissions and correlate them to various levels of C/N. Does higher level of C/N result in better picture and sound quality or higher S/N.
- Measure different levels of S/N by introducing more noise at satellite emulator end and keeping the level of modulating signal constant.
- Correlate the video quality on monitor to different levels of S/N.

RESULT

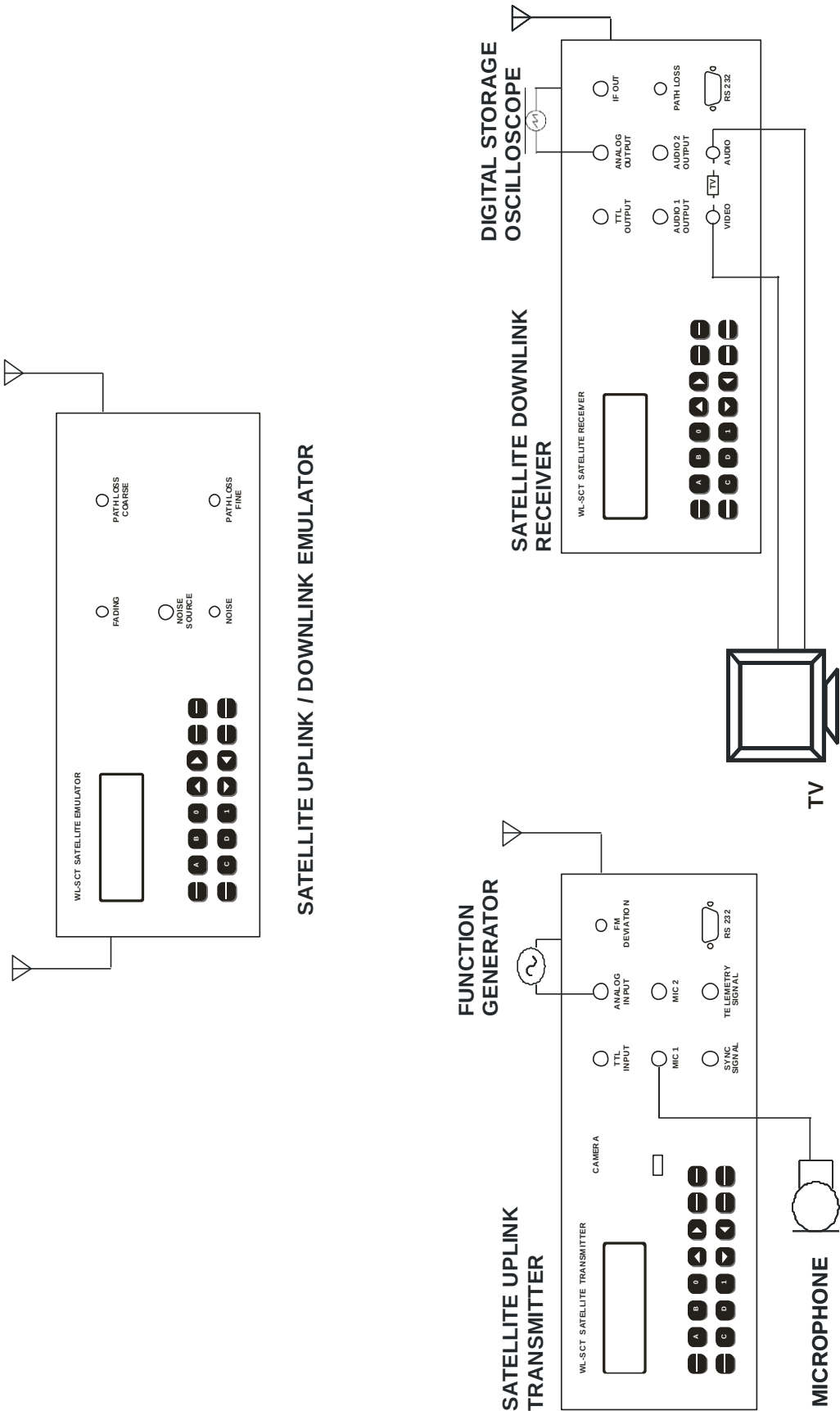
The signal to noise ratio is difference in dB of measured signal level with full modulation and noise floor of the instrument. The actual S/N ratio will depend on a number of parameters at actual link.



**Total signal level i.e.,
signal (peak to peak) + noise**

**Noise level (peak to peak) as read on
OSCILLOSCOPE**

EXPERIMENT NO. 6



Block diagram for Expt.6 Digital baseband signals

EXPERIMENT NO: 6

NAME

Digital Baseband Signals

OBJECTIVE

To measure the digital baseband signal parameters in a satellite communication link and to measure the range of baud rates that the system can support

EQUIPMENTS

- Satellite uplink transmitter, satellite downlink receiver and satellite link emulator
- Antennas
- Function generator, Oscilloscope X 2, spectrum analyzer

PROCEDURE

- To set the Video Link, set the Transmitter & Emulator Uplink Frequency to 2481 MHz, and Receiver & Emulator Downlink frequency to 2400 MHz. This is done to ensure the emulator downlink PLL is locked and displayed frequency is generated correctly.
- If you get the picture on the TV screen at the receiver via satellite, PLL of complete link are O.K. and a successful satellite link is said to be established.
- Set the Function Generator TTL output for 1 KHz.
- Connect the Function Generator O/P to TTL INPUT (which is a digital input, TTL compatible) of Transmitter and connect TTL OUTPUT of Receiver to CRO.
- Set the Input channel VIDEO CH3 of the TRANSMITTER for TTL.

				I	N	P	U	T		C	H	A	N	N	E	L	S		
A	U	D	I	O		C	H	1	:	O	F	F							
A	U	D	I	O		C	H	2	:	O	F	F							
V	I	D	E	O		C	H	3	:	T	T	L							

- Set the Emulators INPUT DATA on VIDEO for digital transmission.

	I	N	P	U	T		D	A	T	A									
A	)		V	I	D	E	O												
B	)		T	E	L	E	M	E	T	R	Y		S	I	G	N	A	L	
C	)		V	I	E	W		R	X		T	E	L		D	A	T	A	

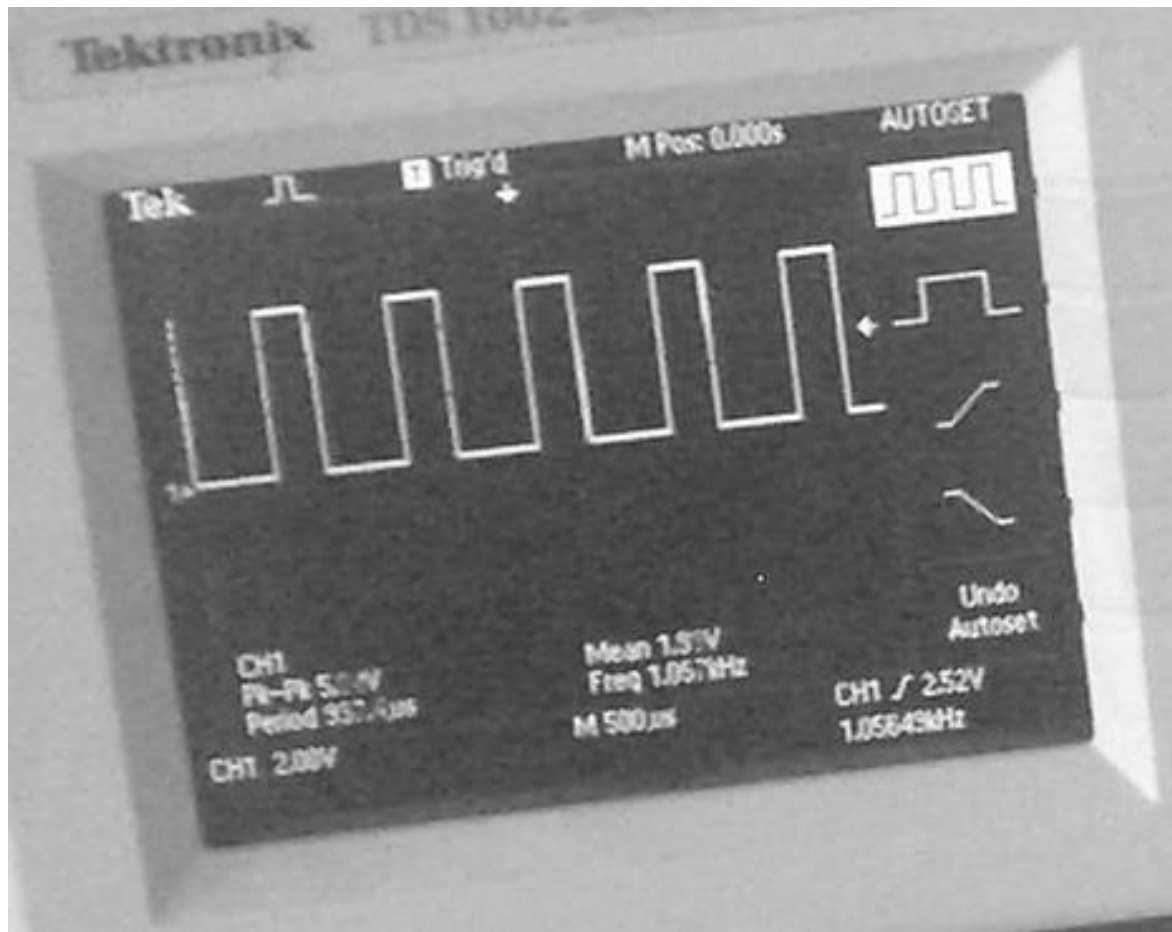
- Similarly Set the Input channel VIDEO CH3 of the RECEIVER for TTL.

				I	N	P	U	T		C	H	A	N	N	E	L	S		
A	U	D	I	O		C	H	1	:	O	F	F							
A	U	D	I	O		C	H	2	:	O	F	F							
V	I	D	E	O		C	H	3	:	T	T	L							

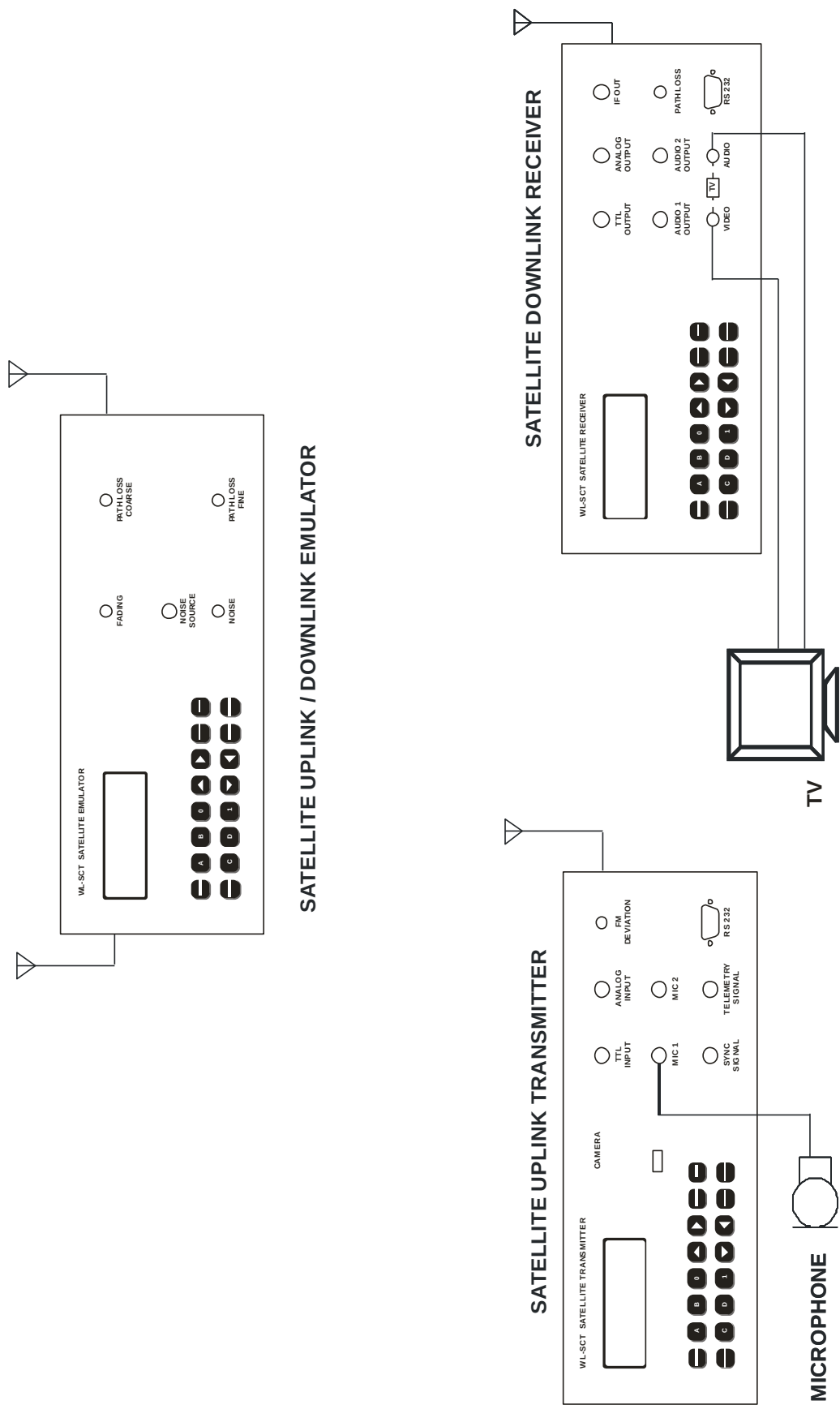
- Preferably connect a DSO at Transmitter end also for viewing transparency of signals using T connectors.
- Now, connect a TTL wave form and Pulse waveforms at TTL in of Transmitter end and vary the frequency of waveforms from 1 KHz to 1 MHz and measure its level, frequency, duty-cycle on DSO. (Use fm deviation pot of TRANSMITTER and fading pot of EMULATOR to adjust the TTL output at receiver)
- Now, measure the level, frequency, duty cycle, noise added to waveform at Receiver end. Measure how much noise or duty cycle variation has been introduced to signal after it has passed through various circuitries of Transmitter, Receiver and transponder (Emulator).
- Send a 50% duty cycle square waveform at Transmitter end. Find the frequency range for which 50% duty cycle can be recovered at Receiver end. Now send a 10% duty cycle wave form and find the frequency range for which the same duty cycle can be recovered. Repeat the same for a 90% duty cycle waveform.
- Is there any difference in frequency range for different duty cycle waveforms? Maximum range of baud rate that system will support is the frequency for which 10-90% waveform can be recovered at Receiver end. This is because information will be transmitted in bursts of data packets in real life.
No video can be sent while data transmission as same video channel is being used for data communication. Audio communication at both channels can take place simultaneously.
- Also find the bandwidth of all the waveforms, which can be supported on digital channel.
- Observe if increasing or decreasing the digital bandwidth from FM deviation control at Transmitter end, has any effect on parameters of received waveforms.
- Observe on DSO, how waveform behaves on fading the carrier by introducing the Fading from satellite link emulator.
- Observe on DSO, how waveform behaves on introducing the noise onto carrier by introducing the noise from satellite link emulator. See if noise introduced can completely shadow the waveforms.

RESULT

The Function Generator TTL O/P waveforms can be transmitted over a distance via a satellite communication link and same TTL O/P waveform can be received at Receiver input. Function Generator digital waveform to be transmitted should vary betweenbps toKbps.



EXPERIMENT NO. 7



Block diagram for Expt.7 Telecommand & Telemetry

EXPERIMENT NO: 7

NAME

Telecommand & Telemetry

OBJECTIVE

To send tele-command and receive the telemetry Data and study the operation of a codec

EQUIPMENTS

- Satellite uplink transmitter, satellite downlink receiver and satellite link emulator
- Antennas
- Function generator, Digital Oscilloscope, spectrum analyzer

PROCEDURE

- To set the Video Link, set the Transmitter & Emulator Uplink Frequency to 2481 MHz, and Receiver & Emulator Downlink frequency to 2400 MHz. This is done to ensure the emulator downlink PLL is locked and displayed frequency is generated correctly.
- If you get the picture on the TV screen at the receiver via satellite, PLL of complete link are O.K. and a successful satellite link is said to be established.
- Set Input channel VIDEO CH3 of Transmitter for TELEMETRY.

				I	N	P	U	T		C	H	A	N	N	E	L	S		
A	U	D	I	O		C	H	1	:	O	F	F							
A	U	D	I	O		C	H	2	:	O	F	F							
V	I	D	E	O		C	H	3	:	T	E	L	E	M	E	T	R	Y	

- After pressing ENTER key, message for feeding DATA and ADDRESS

	T	E	L	E	M	E	T	R	Y		S	E	T	T	I	N	G		
A	)		D	A	T	A			:										
B	)		A	D	D	R	E	S	S	:									

- Use 0 and 1 keys to enter 4-Bit data and 4-Bit Address. Press ENTER key to set the DATA and ADDRESS values.
- Set Input Channels for TELEMETRY SIGNAL in the Emulator.

	C	H	A	N	N	E	L		D	A	T	A							
A	)		V	I	D	E	O												
B	)		T	E	L	E	M	E	T	R	Y								
C	)		V	I	E	W		R	X		T	E	L		D	A	T	A	

- Press key B for setting up TELEMETRY SIGNAL options for the Input channels of the EMULATOR.

	T	E	L	E	M	E	T	R	Y		S	E	T	T	I	N	G	S	
A	)		R	X		T	E	L	E	M	E	T	R	Y					
B	)		T	X		T	E	L	E	M	E	T	R	Y					

Press A key to enter into RX Telemetry

	T	E	L	E	M	E	T	R	Y		S	E	T	T	I	N	G		
A	)		A	D	D	R	E	S	S	:	0	0	0	0					
	E	N	T	E	R		4	B	I	T		A	D	D	R	E	S	S	

- Enter the 4-bit address value using 0 and 1 key. Press ENTER to set the selected address value.
- To check the received data, at emulator press ESC key and then press C key.

					T	E	L	E	M	E	T	R	Y						
S	E	T		A	D	D	R	E	S	S	:				0	0	0	0	
R	E	C	E	I	V	E	D		D	A	T	A	:		0	0	0	0	

- To properly receive the data, adjust the FM deviation pot on the Transmitter. If the selected address is matched, received data will be displayed.
- NOTE:** Make sure Transmitted and received address must be same.
- For setting up the downlink telemetry, press B from Channel data menu.

	C	H	A	N	N	E	L		D	A	T	A							
A	)		V	I	D	E	O												
B	)		T	E	L	E	M	E	T	R	Y								
C	)		V	I	E	W		R	X		T	E	L		D	A	T	A	

Press B key to enter into TX Telemetry

	T	E	L	E	M	E	T	R	Y		S	E	T	T	I	N	G		
A	)		D	A	T	A				:	0	0	0	0					
B	)		A	D	D	R	E	S	S	:	0	0	0	0					
	E	N	T	E	R		4	B	I	T		A	D	D	R	E	S	S	

- Enter the 4-bit address and data value using 0 and 1 key. Press ENTER to set the selected address and data value.
- Set the Input channels VIDEO CH3 of the Receiver for TELEMETRY.

			O	U	T	P	U	T		C	H	A	N	N	E	L	S		
A	U	D	I	O		C	H	1	:	O	F	F							
A	U	D	I	O		C	H	2	:	O	F	F							
V	I	D	E	O		C	H	3	:	T	E	L	E	M	E	T	R	Y	

- Press ENTER to set the selected option for the input channel.

	T	E	L	E	M	E	T	R	Y		S	E	T	T	I	N	G		
A	)			A	D	D	R	E	S	S	:	0	0	0	0				
	E	N	T	E	R			4	B	I	T		A	D	D	R	E	S	S

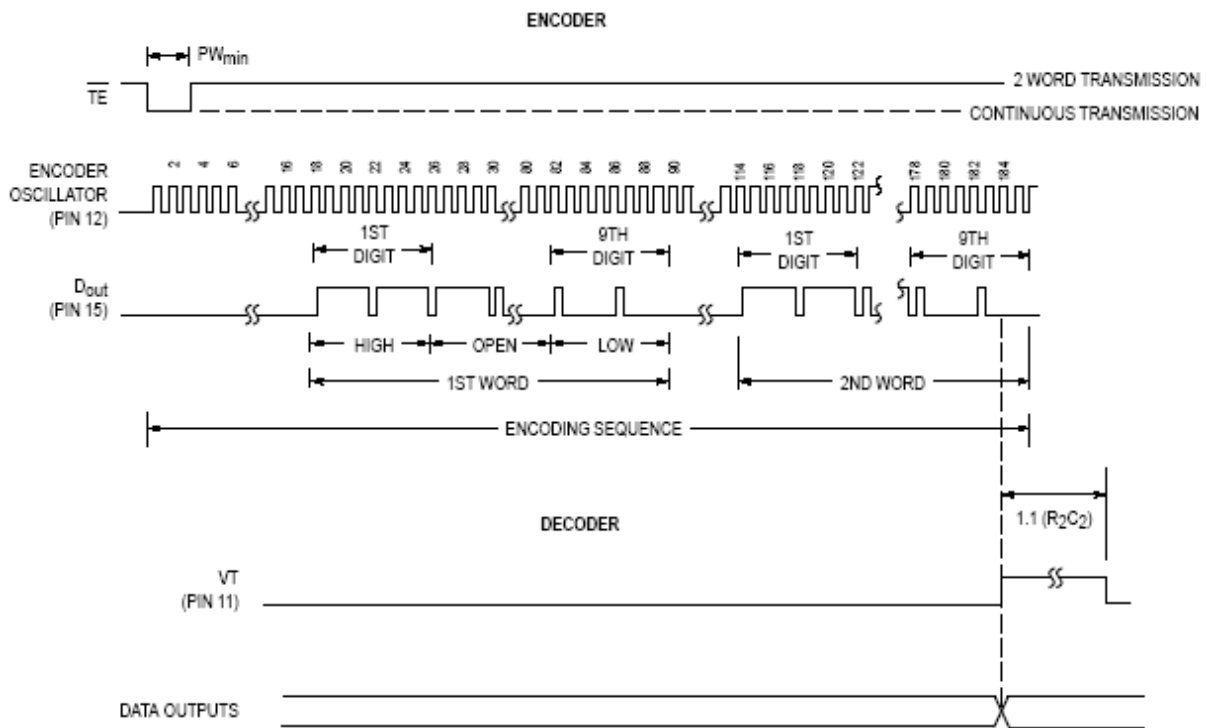
- Enter the 4-bit address value using 0 and 1 key. Use forward arrow key ► and backward arrow key ◀ to move the cursor forward or backward. Press ENTER key to set the selected address value.
If the selected address is matched, received data will be displayed.

	T	E	L	E	M	E	T	R	Y		S	E	T	T	I	N	G		
	S	E	T		A	D	D	R	E	S	S	:	0	0	0	0			
		R	E	C	E	I	V	E	D		D	A	T	A	:	0	0	0	0

- See if noise introduced from the EMULATOR side can completely shadow the data and addresses.

RESULT

The Tele-command and Telemetry signals can be transmitted over a distance via a satellite communication link and same signals can be received at Receiver input. Telecommand function encodes 8 lines of information and serially transmits the information upon receipt of enable signal. The words are transmitted twice per encoding sequence to increase security. The Telemetry function receives the serial data stream and interprets 4 of the digits as address code. The valid led glows on two conditions - first, two addresses must be consecutively received in one encoding sequence, which must match the local addresses. Second the 4 bits of data must match the last 4 bits of valid data received.



Telemetry - Telecommand Data

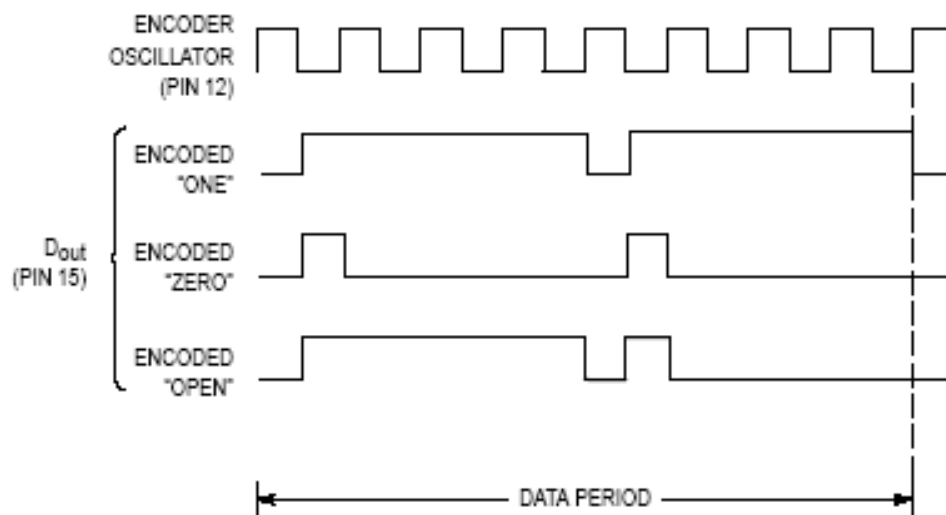
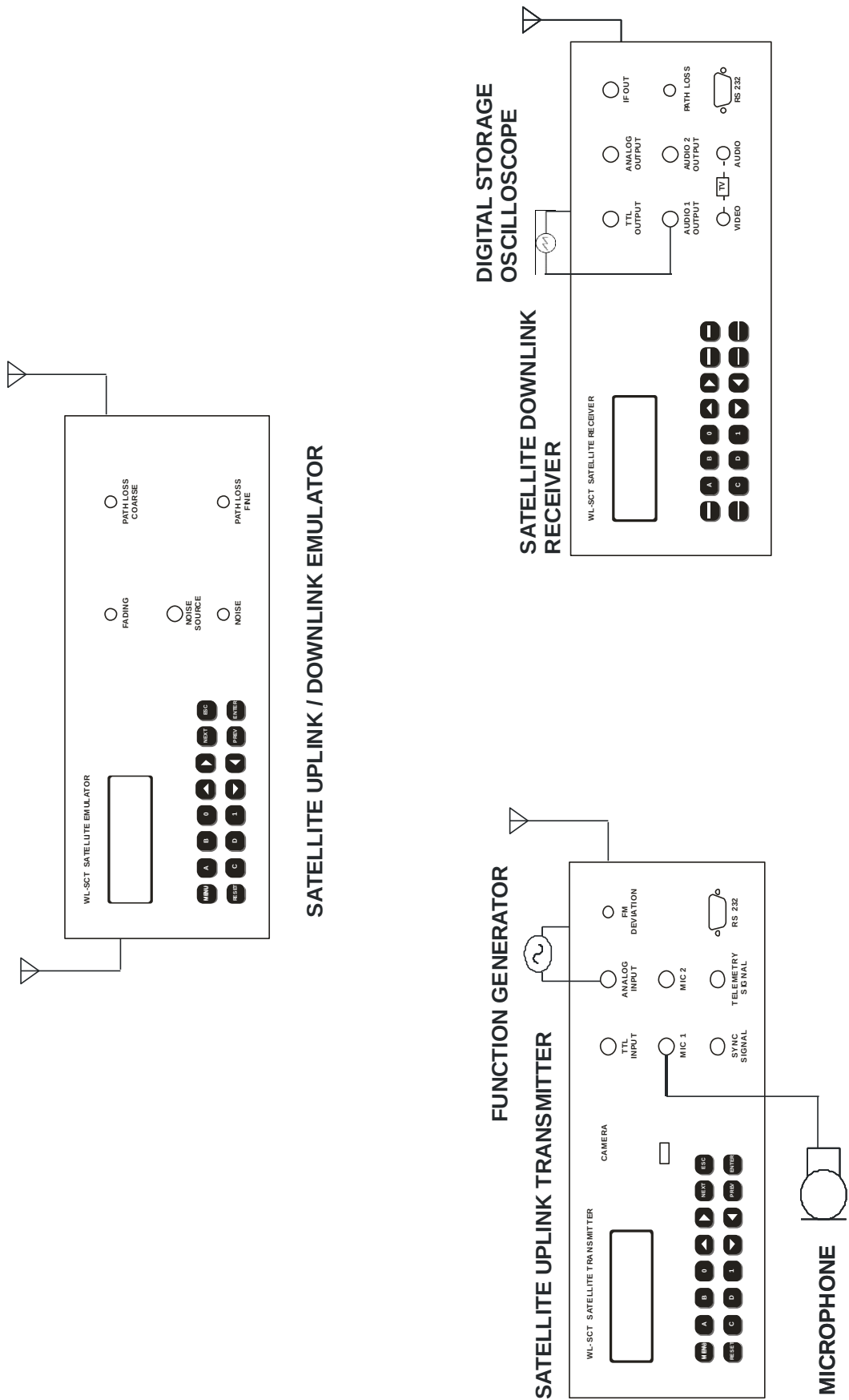


Figure 11. Encoder Data Waveforms

EXPERIMENT NO. 8



Block diagram for Expt.9 Effect of fading and measurement of fading margin

EXPERIMENT NO: 8

NAME

Fading

OBJECTIVE

To study the effect of fading and measure the fading margin of a received signal.

EQUIPMENTS

- Satellite uplink transmitter, satellite downlink receiver and satellite link emulator
- Antennas
- Microphone, video monitor, Function generator, oscilloscope X 2, spectrum analyzer

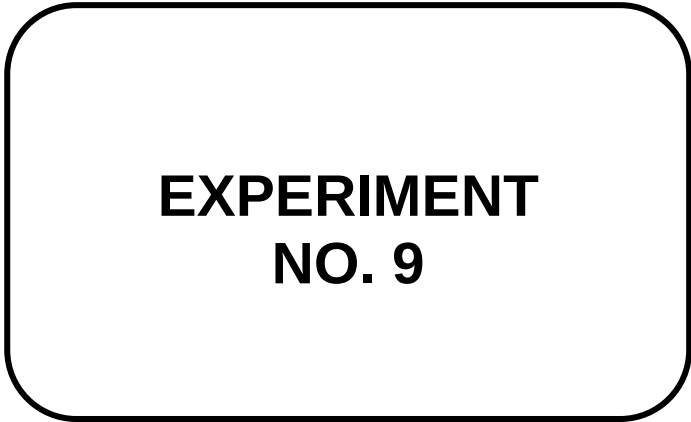
PROCEDURE

- To set the Video Link, set the Transmitter & Emulator Uplink Frequency to 2481 MHz, and Receiver & Emulator Downlink frequency to 2400 MHz. This is done to ensure the emulator downlink PLL is locked and displayed frequency is generated correctly.
- Connect antenna for uplink & downlink transmission & Reception.
- If you get the picture on the TV screen at the receiver via satellite, PLL of complete link are O.K. and a successful satellite link is said to be established.
- Keep Noise, Path Loss Fine potentiometer fully anticlockwise at Emulator & Path Loss at receiver side.
- Keep Fading potentiometer in Emulator fully Anticlockwise & adjust Path Loss Course for complete link.
- Connect 1 KHz 1Vpp sine wave to ANALOG INPUT post of transmitter. set Audio CH1 to Analog in Transmitter and Receiver.
- See if you can receive clearly video as well as audio frequency. Now increase the path loss at both ends and see if you can receive both audio as well as video simultaneously. Why does video signal remain hardly disturbed whereas audio reception is highly susceptible to path loss and multipath effect?
- Observe how does video, audio/sine waves behaves on fading the carrier by introducing the Fading from satellite link emulator.
- Make sure the Receiver is not saturated with carrier otherwise effect of fading might not be visible. This can be done by increasing path loss at Receiver.
- Vary the Fading pot and measure the variation of the carrier level. **Make sure the path loss at satellite down link is high.** Fading can be read as fluctuations in RSSI readings. The difference between maximum and minimum reading of RSSI converted into power level (from chart) will give fading in dB's.
- If received signal strength is reduced to its minimum, one can see the fading in audio and video. **Fading margin** is the variation of carrier allowed during link which doesn't affect the corresponding audio or video.

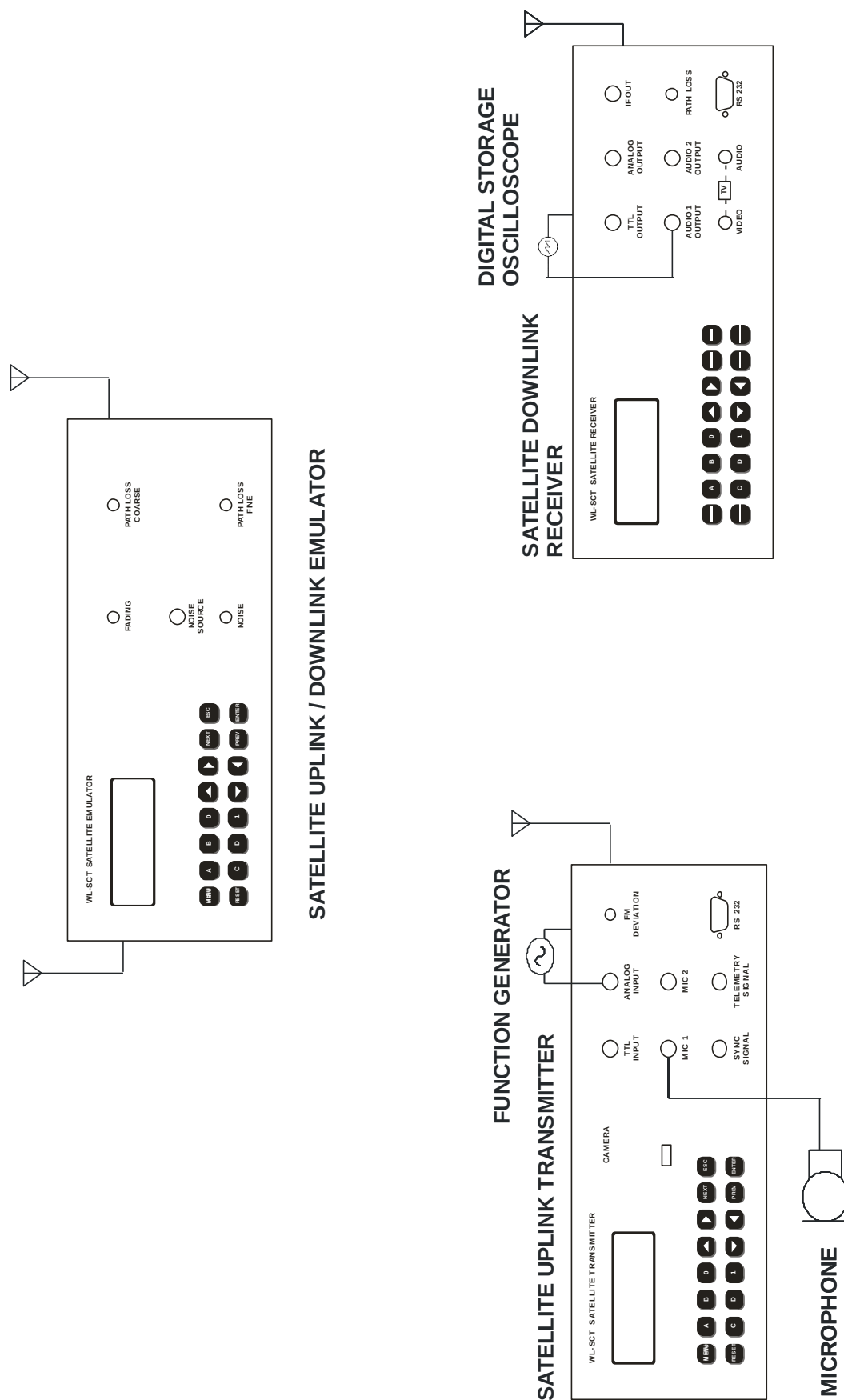
RESULT

Fading is an effect in which carrier level received tends to change with respect to time slowly. Level variation results in changing C/N which can result in a decrease in communication quality. The link may be disrupted entirely if the variation reduces the C/N to below threshold. Enough margin of C/N has to be allocated to allow for fading margins so that no noticeable change is observed in signal.

Fading in video is difficult because of better S/N(because of more bandwidth) but it is much pronounced in audio as audio sub carrier is already 25 dB down from video carrier level plus S/N for audio is much less as compared to video because of little FM deviation allowed.



EXPERIMENT NO. 9



Block diagram for Expt.10 Effect of path loss

EXPERIMENT NO: 9

NAME

Path Loss Effect

OBJECTIVE

To study the effect of path loss and calculate the distance between transmitter & Receiving Antenna

EQUIPMENTS

- Satellite uplink transmitter, satellite downlink receiver and satellite link emulator
- Antennas
- Microphone, video monitor, Function generator, oscilloscope X 2, spectrum analyzer

THEORY

Path loss Concept: To demonstrate the concept of path loss here variable attenuator is used In Emulator & receiver. Path loss is due to the propagation of signal into the space. Received signal strength is indicated on receiver side. As a distance increases free space loss (L) is also increases & reduces the RSSI.

For example:

If the distance between satellite & receiver is 5metre, then

$$\text{Free space loss} = L = 10 n \log_{10} (4\pi D / \lambda)$$

Where

D = the distance between transmitter and receiver antenna.

n = path loss exponent, it is 2 for propagation in free space.

λ = free space wavelength, $C / f = 0.125\text{m}$.

C = Speed of light = $3 \times 10^8 \text{ m/s}$

f = resonance frequency (2.4GHz)

By putting the values in above equation,

$$L = 10 (2) \log_{10} (4\pi(5) / (0.125))$$

$$L = 54.02\text{dB}$$

PROCEDURE

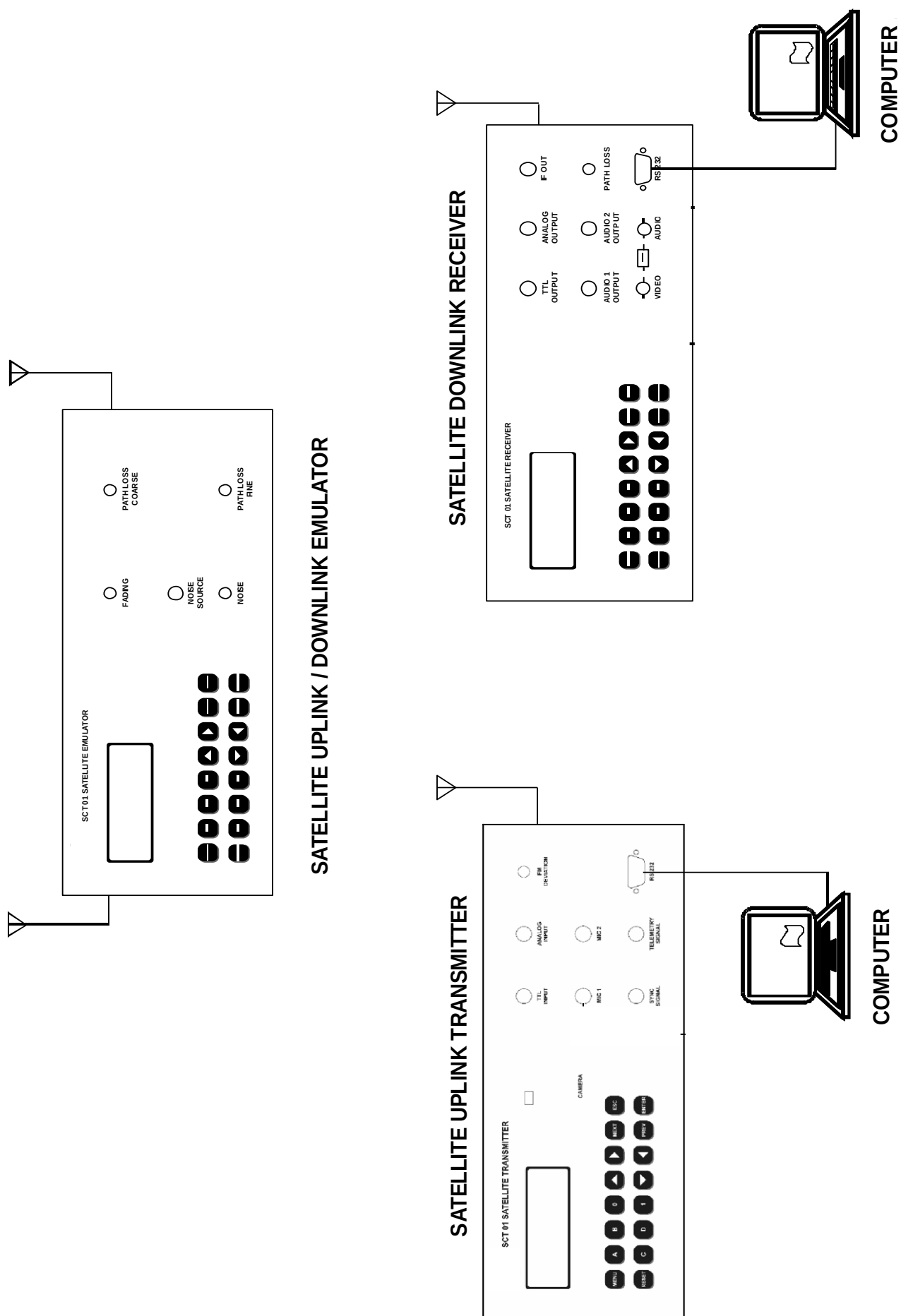
- To set the Video Link, set the Transmitter & Emulator Uplink Frequency to 2481 MHz, and Receiver & Emulator Downlink frequency to 2400 MHz. This is done to ensure the emulator downlink PLL is locked and displayed frequency is generated correctly.
- Connect antenna in uplink (i. e to transmitter and Rx post of Emulator). Connect antenna in Down link (i. e to Tx post of emulator and to receiver)
- If you get the picture on the TV screen at the receiver via satellite, PLL of complete link are O.K. and a successful satellite link is said to be established.

- Keep Noise, Path Loss Fine potentiometer fully anticlockwise at Emulator & Path Loss at receiver side.
- Keep Fading potentiometer in Emulator fully Anticlockwise & adjust Path Loss Course for complete link.
- Connect 1 KHz 1Vpp sine wave to ANALOG INPUT post of transmitter. Set Audio CH1 to Analog in Transmitter and Receiver.
- Observe the demodulated analog signal at AUDIO 1 OUTPUT post of receiver on CRO and observe Video signal on TV.
- Now increase the attenuation using Path Loss pot provided on emulator & receiver side & observe the corresponding effect on audio & video channel.
go on increasing the attenuation till received audio & video signal gets disturbed, note the RSSI reading and calculate the corresponding distance between transmitter & Receiving Antenna.
- Change the uplink and downlink frequency and repeat the procedure from steps 4 to 8.
- To calculate lossy media, take value of path loss exponent $n = 4$. Increase the noise using noise pot on emulator and do the procedure from 4 to 8.

RESULT

As a distance between Transmitter and Receiver increases, Path loss increases. Received signal strength at receiver is inversely proportional to Path loss. As a path loss increases, received signal strength at receiver decreases. This results in a distortion in received signal.

**EXPERIMENT
NO. 10**



EXPERIMENT NO:10

NAME

PC-PC Communication with Rs-232 using Satellite Communication Link

OBJECTIVE

To setup a RS-232 satellite communication link using RS 232 ports.

EQUIPMENTS

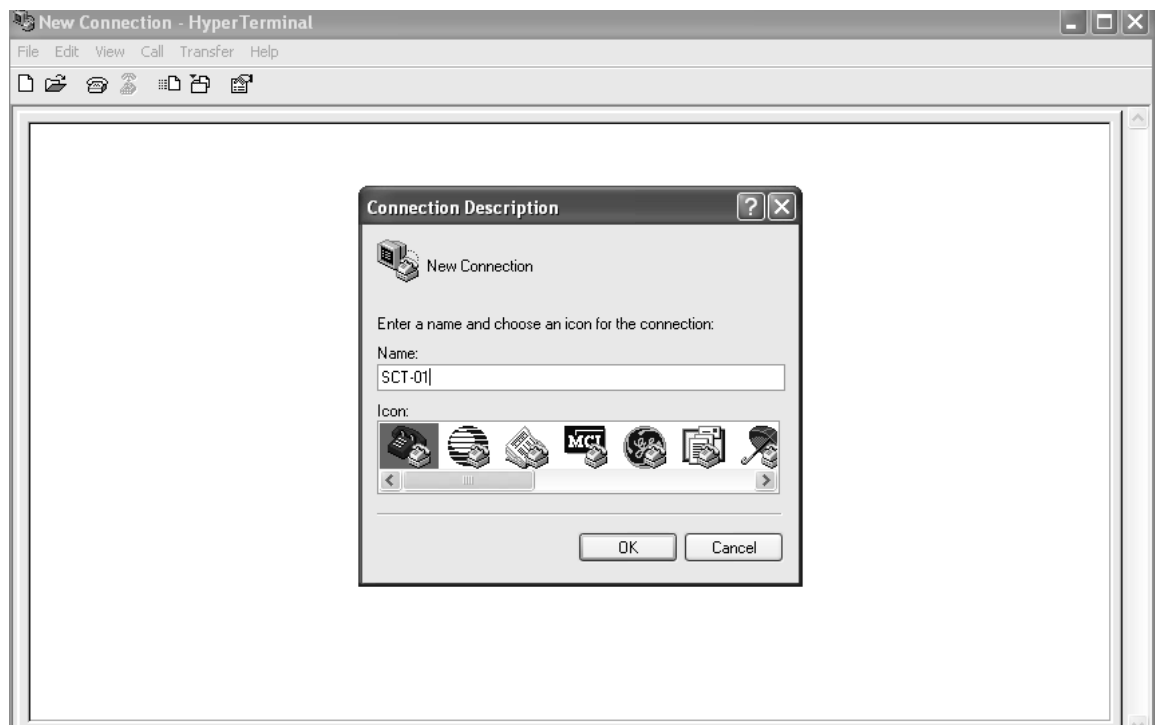
- Satellite uplink transmitter, satellite downlink receiver and satellite link emulator
- Helix antennas
- Computer-2 Nos.

PROCEDURE

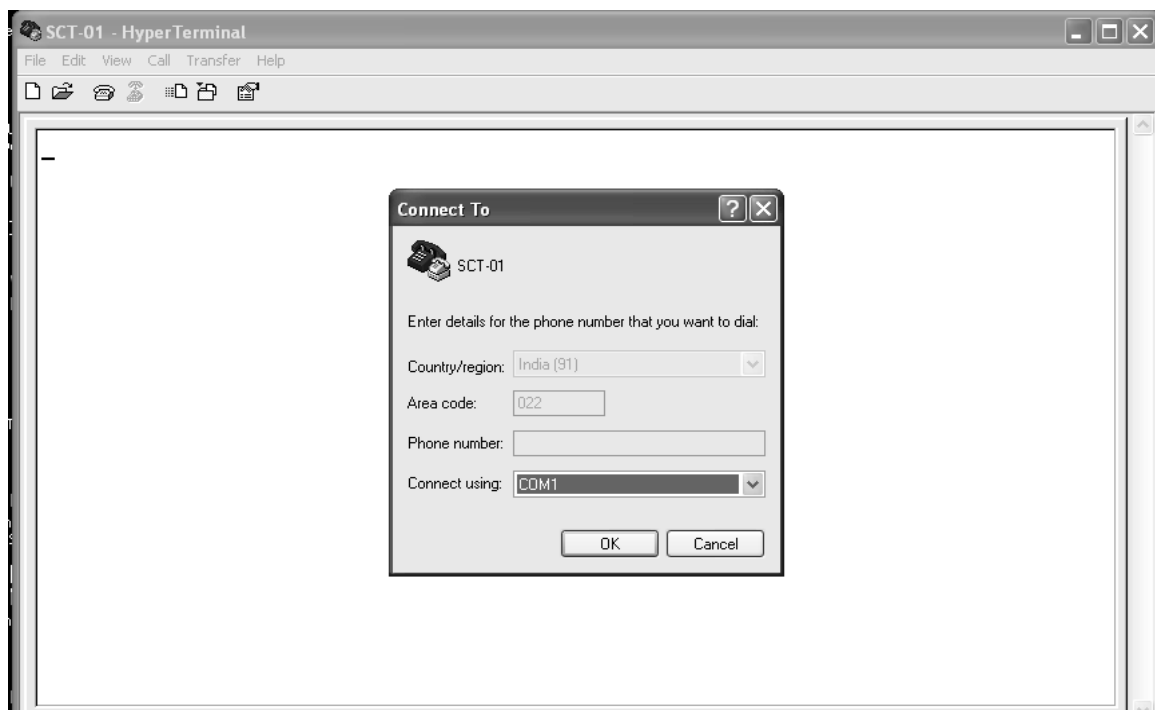
- To set the Video Link, set the Transmitter & Emulator Uplink Frequency to 2481 MHz, and Receiver & Emulator Downlink frequency to 2400 MHz. This is done to ensure the emulator downlink PLL is locked and displayed frequency is generated correctly.
- If you get the picture on the TV screen at the receiver via satellite, PLL of complete link are O.K. and a successful satellite link is said to be established.
- Set the Input channel VIDEO CH3 of the Satellite Transmitter for RS-232. Move the cursor on display up or down using up key ▲ and down key ▼. Use forward ► and backward ◀ arrow key to select the input channel option. Press ENTER to set the input channel selection.

				I	N	P	U	T		C	H	A	N	N	E	L	S		
A	U	D	I	O		C	H	1	:	O	F	F							
A	U	D	I	O		C	H	2	:	O	F	F							
V	I	D	E	O		C	H	3	:	R	S	2	3	2					

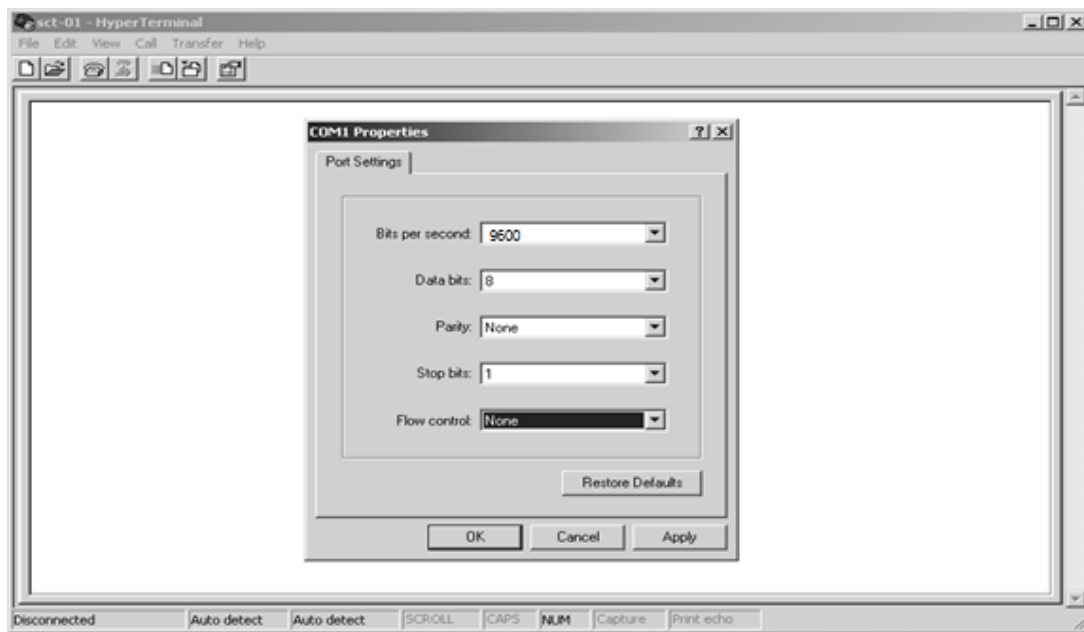
- Connect RS-232 port from front panel of transmitter to COM port of Computer.
- Open the hyper terminal Utility from the Program / Accessories / Communications / Hyper Terminal. Following Window will appear.



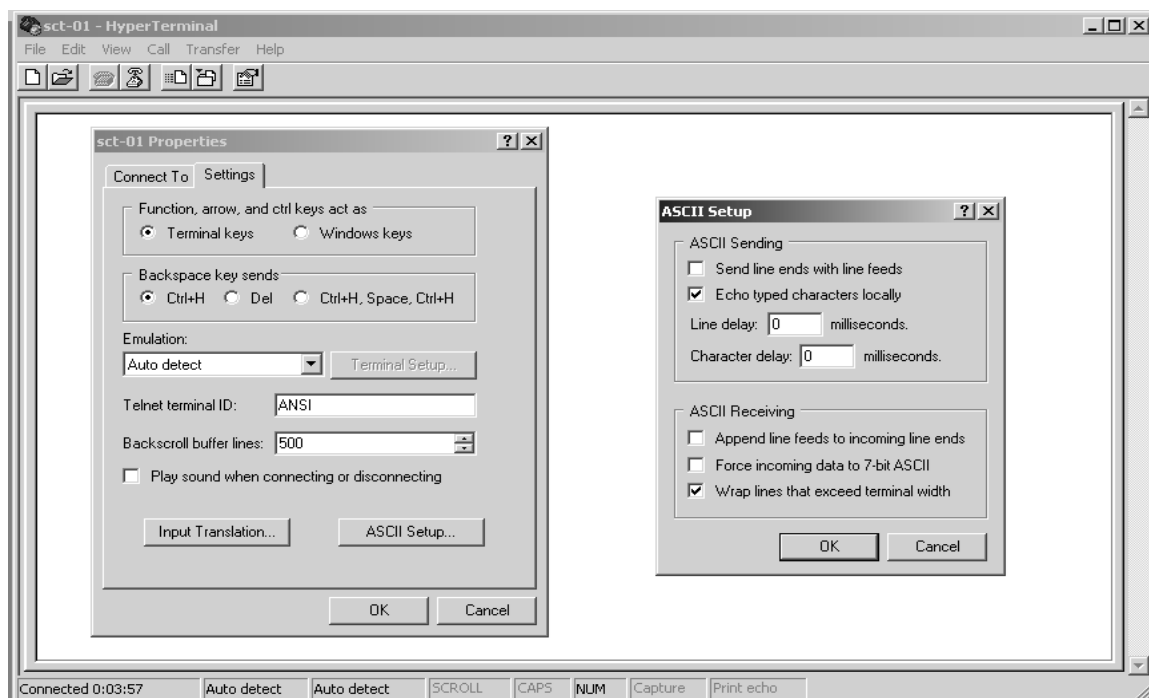
- Press OK to continue. Select COM port from following window and Press OK



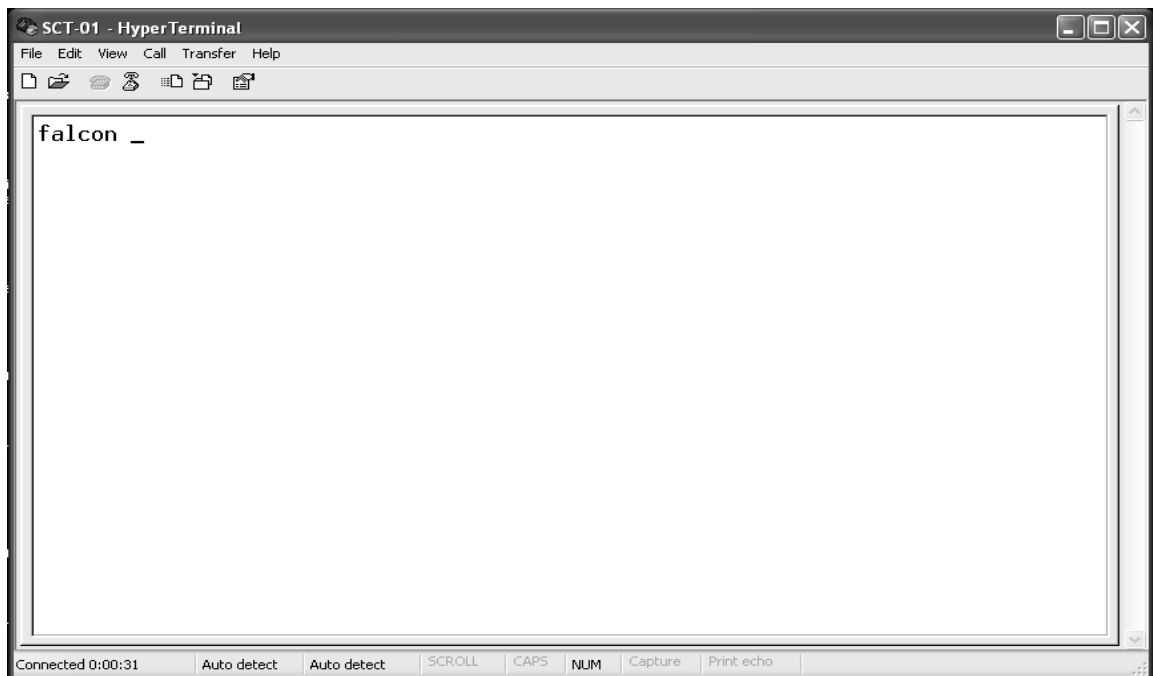
- Select Baud rate as 9600, Data bits 8, Parity None, Stop bit 1 and Flow Control None. Press Apply and the OK to save your settings.



- For next setting disconnect the hyper terminal connection Click on properties icon Go to setting tab then go to ASCII Setup and check mark on Echo typed characters locally Click ok to save the setting.



- Once you Set the settings as per above settings following window will appear where you can type the message to sent using communication link.



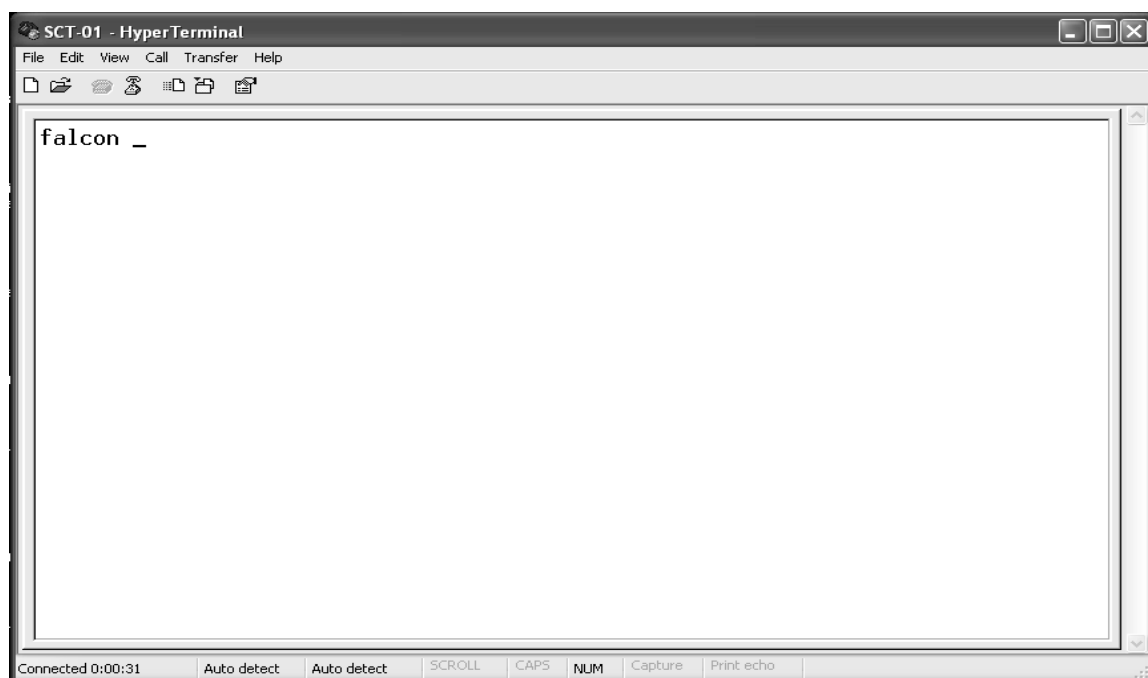
- Press A key to set the Emulators INPUT DATA on VIDEO for digital transmission.

	I	N	P	U	T		D	A	T	A									
	A	)		V	I	D	E	O											
	B	)		T	E	L	E	M	E	T	R	Y		S	I	G	N	A	L
	C	)		V	I	E	W		R	X		T	E	L		D	A	T	A

- Similarly Set the Input channel VIDEO CH3 of the RECEIVER for RS-232.

			O	U	T	P	U	T		C	H	A	N	N	E	L	S			
A	U	D	I	O		C	H	1	:	O	F	F								
A	U	D	I	O		C	H	2	:	O	F	F								
V	I	D	E	O		C	H	3	:	R	S	2	3	2						

- Connect RS-232 port from front panel of Receiver to COM port of other Computer.
- Set the same hyper terminal settings as per transmitter on receiver Computer.
- Keep all path loss POT on emulator and receiver to fully anti-clockwise direction. Keep Noise POT on emulator to fully anti-clockwise direction. Keep Fading Pot on Emulator to fully anti-clockwise direction.
- Now you can start receiving the characters on receiver computer.



POWER CONVERSION TABLE 75 OHMS SYSTEM

dBm	pW	dBmV	dBμV	μV_{RMS}	μV_{Peak}
-110	0.01	-61.2	-1.2	.866	1.22
-109	0.012	-60.2	-0.2	.97	1.37
-108	0.016	-59.2	0.8	1.09	1.54
-107	0.020	-58.2	1.8	1.22	1.73
-106	0.025	-57.2	2.8	1.37	1.94
-105	0.031	-56.2	3.8	1.54	2.18
-104	0.040	-55.2	4.8	1.79	2.44
-103	0.050	-54.2	5.8	1.93	2.74
-102	0.063	-53.2	6.8	2.17	3.07
-101	0.079	-52.2	7.8	2.44	3.45
-100	0.100	-51.2	8.8	2.74	3.87
-99	0.126	-50.2	9.8	3.07	4.35
-98	0.158	-49.2	10.8	3.45	4.88
-97	0.198	-48.2	11.8	3.87	5.47
-96	0.251	-47.2	12.8	4.34	6.14
-95	0.316	-46.2	13.8	4.87	6.89
-94	0.39	-45.2	14.8	5.46	7.73
-93	0.50	-44.2	15.8	6.13	8.67
-92	0.63	-43.2	16.8	6.88	9.73
-91	0.79	-42.2	17.8	7.72	10.9
-90	1.00	-41.2	18.8	8.66	12.2
-89	1.26	-40.2	19.8	9.7	13.7
-88	1.58	-39.2	20.8	10.9	15.4
-87	1.99	-38.2	21.8	12.2	17.3
-86	2.51	-37.2	22.8	13.7	19.4
-85	3.16	-36.2	23.8	15.4	21.8

dBm	pW	dBmV	dBμV	μV_{RMS}	μV_{Peak}
-84	3.98	-35.2	24.8	17.9	24.4
-83	5.01	-34.2	25.8	19.3	27.4
-82	6.31	-33.2	26.8	21.7	30.7
-81	7.9	-32.2	27.8	24.4	34.5
-80	10.0	-31.2	28.8	27.4	38.7
-79	12.6	-30.2	29.8	30.7	43.5
-78	15.8	-29.2	30.8	34.5	48.8
-77	19.9	-28.2	31.8	38.7	54.7
-76	25.1	-27.2	32.8	43.4	61.4
-75	31.6	-26.2	33.8	48.7	68.9
-74	39.8	-25.2	34.8	54.6	77.3
-73	50.1	-24.2	35.8	61.3	86.7
-72	63.1	-23.2	36.8	68.8	97.3
-71	79.0	-22.2	37.8	77.2	109
-70	100	-21.2	38.8	86.6	122
-69	126	-20.2	39.8	97	137
-68	158	-19.2	40.8	109	154
-67	199	-18.2	41.8	122	173
-66	251	-17.2	42.8	137	194
-65	316	-16.2	43.8	154	218
-64	398	-15.2	44.8	179	244
-63	501	-14.2	45.8	193	274
-62	631	-13.2	46.8	217	307
-61	790	-12.2	47.8	244	345
-60	1000	-11.2	48.8	274	387
-59	1260	-10.2	49.8	307	435
-58	1580	-9.2	50.8	345	488
-57	1980	-8.2	51.8	387	547
-56	2510	-7.2	52.8	434	614

dBm	nW	dBmV	dBμV	mV_{rms}	mV_{peak}
-55	3.16	-6.2	53.8	.487	.689
-54	3.98	-5.2	54.8	.546	.773
-53	5.01	-4.2	55.8	0.613	0.867
-52	6.31	-3.2	56.8	0.688	0.973
-51	7.9	-2.2	57.8	0.772	1.09
-50	10.0	-1.2	58.8	0.866	1.22
-49	12.6	-0.2	59.8	0.97	1.37
-48	15.8	0.8	60.8	1.09	1.54
-47	19.9	1.8	61.8	1.22	1.73
-46	25.1	2.8	62.8	1.37	1.94
-45	31.6	3.8	63.8	1.54	2.18
-44	39.8	4.8	64.8	1.79	2.44
-43	50.1	5.8	65.8	1.93	2.74
-42	63	6.8	66.8	2.17	3.07
-41	79	7.8	67.8	2.44	3.45
-40	100	8.8	68.8	2.74	3.87
-39	126	9.8	69.8	3.07	4.35
-38	158	10.8	70.8	3.45	4.88
-37	198	11.8	71.8	3.87	5.47
-36	251	12.8	72.8	4.34	6.14
-35	316	13.8	73.8	4.87	6.89
-34	398	14.8	74.8	5.46	7.73
-33	501	15.8	75.8	6.13	8.67
-32	631	16.8	76.8	6.88	9.73
-31	790	17.8	77.8	7.72	10.9
-30	1.00	18.8	78.8	8.66	12.2
-29	1.26	19.8	79.8	9.7	13.7
-28	1.58	20.8	80.8	10.9	15.4
-27	1.99	21.8	81.8	12.2	17.3
-26	2.51	22.8	82.8	13.7	19.4
-25	3.16	23.8	83.8	15.4	21.8

dBm	nW	dBmV	dBμV	mV_{rms}	mV_{peak}
-24	3.98	24.8	84.8	17.9	24.4
-23	5.01	25.8	85.8	19.4	27.4
-22	6.34	26.8	86.8	21.7	30.7
-21	7.94	27.8	87.8	24.4	34.5
-20	10.0	28.8	88.8	27.4	38.7
-19	12.6	29.8	89.8	30.7	43.5
-18	15.8	30.8	90.8	34.5	48.8
-17	19.9	31.8	91.8	38.7	54.7
-16	25.1	32.8	92.8	43.4	61.4
-15	31.6	33.8	93.8	48.7	68.9
-14	39.8	34.8	94.8	54.6	77.3
-13	50.1	35.8	95.8	61.3	86.7
-12	63.1	36.8	96.8	68.8	97.3
-11	79.4	37.8	97.8	77.2	109.2
-10	100	38.8	98.8	86.6	122.5
-9	126	39.8	99.8	97.2	137.5
-8	158	40.	100.8	108.9	154.0
-7	199	41.8	101.8	122.2	172.8
-6	251	42.8	102.8	137.2	194
-5	316	43.8	103.8	154.0	217.8
-4	398	44.8	104.8	172.8	244.4
-3	501	45.8	105.8	193.8	274.0
-2	631	46.8	106.8	217.5	307.6
-1	0.794	47.8	107.8	244.0	345.0
0	1.000	48.8	108.8	273.9	387.3
+1	1.259	49.8	109.8	307.3	434.6
+2	1.585	50.8	110.8	344.8	487.6
+3	1.995	51.8	111.8	386.8	547.1
+4	2.512	52.8	112.8	434.0	613.8
+5	3.162	53.8	113.8	487.0	688.7
+6	3.981	54.8	114.8	546.4	772.8
+7	5.012	55.8	115.8	613.1	867.1

dBm	nW	dBmV	dBμV	mV_{rms}	mV_{peak}
+8	6.310	56.8	116.8	687.9	972.8
+9	7.943	57.8	117.8	771.8	1091.6
+10	10.000	58.8	118.8	866.0	1224.7

POWER – RSSI (Satellite Receiver) CONVERSION TABLE 75 OHMS SYSTEM:-

dBm	pW	RSSI (V)
-96	0.251	0.92
-95	0.361	0.93
-94	0.39	0.93
-93	0.50	0.93
-92	0.63	0.94
-91	0.79	0.95
-90	1.00	0.97
-89	1.26	0.99
-88	1.58	1.01
-87	1.99	1.03
-86	2.51	1.06
-85	3.16	1.08
-84	3.98	1.1
-83	5.01	1.13
-82	6.31	1.16
-81	7.9	1.18
-80	10.0	1.21
-79	12.6	1.24
-78	15.8	1.27
-77	19.9	1.30
-76	25.1	1.33
-75	31.6	1.37
-74	39.8	1.40
-73	50.1	1.43
-72	63.1	1.46
-71	79.0	1.51
-70	100	1.54
-69	126	1.58
-68	158	1.62
-67	199	1.65

dBm	nW	RSSI(V)
-66	251	1.69
-65	316	1.73
-64	398	1.76
-63	501	1.80
-62	631	1.83
-61	790	1.87
-60	1000	1.90
-59	1260	1.93
-58	1580	1.97
-57	1980	2.01
-56	2510	2.51
-55	3.16	3.06
-54	3.98	3.34
-53	5.01	3.49
-52	6.31	3.58
-51	7.9	3.63
-50	10.0	3.67
-49	12.6	3.70
-48	15.8	3.72
-47	19.9	3.74
-46	25.1	3.75
-45	31.6	3.77
-44	39.8	3.78
-43	50.1	3.80
-42	63	3.80
-41	79	3.81
-40	100	3.82
-39	126	3.83
-38	158	3.84
-37	198	3.86
-36	251	3.88

dBm	nW	RSSI(V)
-35	316	3.92
-34	398	4.00
-33	501	4.06
-32	631	4.1
-31	790	4.13
-30	1000	4.14
-29	1260	4.17
-28	1580	4.18
-27	1990	4.20
-26	2510	4.21
-25	3160	4.22
-24	3980	4.22
-23	5010	4.23
-22	6340	4.24
-21	7940	4.25
-20	10,000	4.25